Estimating Choice Models with Unobserved Expectations over Attributes

Mathias Reynaert Wenxuan Xu Hanlin Zhao *

March 10, 2026

Abstract

Agents often make choices based on expectations about uncertain attributes, yet the information used to form these expectations is typically unobserved by researchers. We develop two methods to estimate discrete choice models in which agents form expectations using heterogeneous and unobserved information sets. Preferences are point-identified by approximating the latent information structure with a finite mixture of information types, or set-identified with partial information about expectations. Both approaches apply to individual- and market-level data with minimal assumptions on expectation formation. Applications to firms' exporting decisions and consumers' automobile purchases show that accounting for unobserved information substantially affects estimated preferences.

Keywords: discrete choice, unobserved information, mixture model, set identification
JEL codes: C5, C8, D8

*Reynaert: Toulouse School of Economics, University of Toulouse Capitole and CEPR, Xu: National University of Singapore, and Zhao: Toulouse School of Economics, University of Toulouse Capitole. We thank Christian Bontemps, Zach Brown, Chris Conlon, Michael Dickstein, Arnaud Maurel and seminar participants at Toulouse School of Economics, Vienna University, APIOC 2024, IIOC 2025 and EARIE 2025 for useful discussions. We acknowledge funding by the European Union (ERC, SPACETIME, grant n° 101077168). Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union or the European Research Council Executive Agency. Neither the European Union nor the granting authority can be held responsible for them. We acknowledge funding from ANR under grant ANR-17-EURE-0010 (Investissements d'Avenir program).

1 Introduction

Many economic choices depend on expectations about uncertain attributes. Consumers anticipate future fuel costs when purchasing vehicles (Hausman, 1979), individuals form expectations about their health risk when selecting insurance (Einav et al., 2010), and students forecast future wages when choosing educational pathways (Arcidiacono et al., 2020). All these settings require the researcher to account for the expectations agents form about the uncertain attributes of their options in order to estimate preferences.

Modeling expectations is challenging because researchers rarely observe the information sets agents use to predict uncertain attributes. Moreover, agents are likely to differ in the information they use. Understanding how heterogeneous information shapes decisions is central to evaluating information policies and regulation in markets with uncertainty. Most empirical applications impose a specific expectation formation process—often perfect foresight—and plug the resulting predictions into the choice model. When the researcher's assumed prediction rule differs from the agents' actual information structure, preference estimates are generally biased (Manski, 1993, 2004; Cunha and Heckman, 2007; Dickstein and Morales, 2018). The central identification challenge is that observed variation in choices may reflect either heterogeneous preferences or heterogeneous beliefs about uncertain attributes.

This paper develops two complementary approaches to address unobserved and heterogeneous information sets in discrete choice models with individual- or market-level data. The first delivers point identification by modeling information heterogeneity as a finite mixture over prediction rules, allowing the researcher to recover the distribution of information types. The second achieves set identification under weaker assumptions by deriving moment inequalities that bound preferences when the information structure is only partially observed. Together, these methods provide a formal identification framework for discrete choice models with unobserved information heterogeneity.

Our first approach generalizes standard plug-in methods that impose a single expectation formation rule. We model agent expectations as a finite mixture over heterogeneous information types. Specifically, we specify observable and latent information variables used to predict the uncertain attribute and construct combinations of these variables as candidate information types. For each type, we form the corresponding conditional expectation under rational expectations (Ahn and Manski, 1993) and incorporate these type-specific predictions into the choice model. Estimating the model as a finite mixture recovers the distribution of information types from choice data. Relative to Cunha et al.

(2005) (CHN05 hereafter), who assume a common information set and focus on selection, our framework allows for heterogeneous information sets and identifies both preferences and the distribution of information types.[1]

We establish point identification of preferences when the finite set of candidate types spans the information sets present in the data-generating process. The key requirement is that the researcher specifies a sufficiently rich set of observed and latent information candidates governing the uncertain attribute so that the induced conditional expectations span the support of agents' prediction rules and thereby restrict the support of belief heterogeneity. This yields a finite-mixture representation with known support. This structure allows us to derive explicit rank conditions for identification and to embed the model within the general random utility framework of Berry and Haile (2014), extending their identification results to settings with unobserved information heterogeneity. This provides a formal characterization of when plug-in approaches deliver consistent preference estimates and when they do not. We further show how the framework accommodates market-specific information distributions, diagnostic tests of the information structure, and establish identification in models with preference heterogeneity and endogenous product characteristics.

We develop a second approach that requires observing only a single information type and imposes a weaker structure on the information environment. In this setting, preferences are set identified rather than point identified. We formally characterize the source of non-identification, showing that it arises from unrestricted dispersion in prediction errors across information types. We derive moment inequalities that bound preferences by exploiting realizations in which the observed type attains extremal prediction errors, so that its choice probabilities bound those of other types, in the spirit of D'Haultfœuille et al. (2018). Unlike the finite-mixture approach, this method does not require enumerating all information sets and does not rely on a minimal-information assumption as in Dickstein and Morales (2018) (DM18 hereafter). We then characterize the identifying power of these inequalities theoretically and in simulations.

Finally, we apply our methods to two empirical settings. First, revisiting DM18, we estimate a finite-mixture model of exporters' profit expectations using observed ex-post profits. The mixture approach recovers the distribution of information types which reveals how firms differ in the information they use to form expectations. We find preference estimates consistent with DM18's set-identification results. We also implement our moment-

[1] As in Dickstein and Morales (2018), we observe the ex-post realization of the uncertain attribute for all alternatives. In contrast to Cunha et al. (2005), where the outcome is observed only for the chosen alternative, we therefore do not face a selection problem when estimating conditional expectations.

inequality method in this setting and obtain comparable bounds.

In a second application, we revisit the estimation of fuel cost valuations in automobile purchases (Grigolon et al., 2018). The existing literature typically assumes perfect foresight by plugging realized fuel costs into the choice model. Applying our finite-mixture approach, we estimate heterogeneous expectation rules based on combinations of observable fuel-related information. We find that the standard plug-in approach substantially overstates consumers' valuation of fuel costs, leading to economically meaningful distortions in estimated fuel cost sensitivity, and that full-information consumers can be rejected. Instead, most buyers appear to rely on simplified prediction rules based on fuel efficiency or fuel type rather than combining both sources of information.

Our paper contributes to the empirical literature on beliefs and expectations in discrete choice. Relative to DM18 and its extensions (Dickstein et al., 2024; Porcher et al., 2024), we adopt a different information structure that permits point identification through a finite-mixture representation and yields moment inequalities compatible with market-level data.

Our finite-mixture approach generalizes earlier rational-expectations frameworks such as CHN05 and Cunha and Heckman (2007), with recent applications in Houmark et al. (2024) and Aucejo and James (2021). These models specify how information enters expectations and test alternative information sets under a common information structure. In contrast, we estimate a distribution over information sets within a unified framework. If the data-generating process features a single homogeneous information set, the mixture collapses to that type; otherwise, it recovers heterogeneous information structures. Recovering the distribution of information types is particularly relevant for policy analysis, where interventions can be information-based and require disentangling preference heterogeneity from information heterogeneity (Allcott and Knittel, 2019; Barahona et al., 2023). The finite-mixture framework accommodates multinomial choice models, market-level data, unobserved preference heterogeneity, and endogenous characteristics.

This paper develops tools to separate preference heterogeneity from belief heterogeneity in environments where expectations are formed under rational but unobserved information structures. Other approaches rely on direct measures of beliefs or information formation, such as surveys, experiments, or quality signals (Arcidiacono et al., 2020; Brown and Jeon, 2024; Vatter, 2024). In contrast, we require only the realization of the uncertain attribute and a set of observable information variables used to form expectations. While Abaluck and Compiani (2020) show that choice data alone can identify preferences under uncertainty, our framework instead exploits structured information variables to recover

both preferences and the distribution of information types. More broadly, our approach complements work on decision-making with latent information (Bergemann et al., 2022) by providing identification results in environments where beliefs are rational but information sets are heterogeneous and unobserved.

Our paper is structured as follows. Section 2 presents the choice model. Section 3 explains the identification issues stemming from the unobserved information structure. Section 4 introduces the finite mixture approach. Section 5 introduces the moment inequality approach. Section 6 presents the simulation results. Section 7 presents the two empirical applications.

2 Choice Model

There are T markets indexed by $t \in \mathcal{T} = \{1, \ldots, T\}$. In each market t there are N_t individuals indexed by $i \in \mathcal{N}_t = \{1, \ldots, N_t\}$. Each individual i chooses one option $j \in \mathcal{J} = \{0, 1, \ldots, J\}$ where $j = 0$ denotes the outside option. We assume that the (realized) utility that individual i derives from option j in market t is given by

$$u_{ijt} \equiv X_{ijt}'\beta + \gamma g_{ijt} + \epsilon_{ijt}, \tag{1}$$

where $X_{ijt} \in \mathbb{R}^{|X|}$ is a vector of observed attributes, $g_{ijt} \in \mathbb{R}$ denotes the realized value of an attribute that is uncertain to individuals at the time of decision, $\epsilon_{ijt} \in \mathbb{R}$ is an idiosyncratic taste shock that is i.i.d. across options j, conditional on (i, t). The researcher observes (X_{ijt}, g_{ijt}).[2] The outside option utility is normalized to $u_{i0t} = \epsilon_{i0t}$. The vector of preference parameters of interest is $\theta = (\beta', \gamma)'$. We begin by presenting the model under homogeneous preferences and exogenous characteristics. Section 4.3 extends the framework to allow for preference heterogeneity and endogenous attributes.

Let $\mathcal{I}_{it}$ denote individual i's information set in market t which includes known attributes, the primitives, and signals about g_{ijt}. We assume that individuals use a subset of this information to form predictions about the uncertain attribute g_{ijt}. Let $\mathcal{W}_{it} \subseteq \mathcal{I}_{it}$ denote the collection of information variables used by individual i to predict g_{ijt}. We impose rational expectations, so that,

$$g_{ijt}^e \equiv \mathbb{E}\left[g_{ijt} \mid \mathcal{W}_{it}\right].$$

[2] As in DM18, the researcher observes the realization of g_{ijt} for all j. This differs from settings with selection (as in CHN05), where g_{ijt} is only observed for the chosen option.

Expected utility is:

$$u^e_{ijt} \equiv X'_{ijt}\beta + \gamma g^e_{ijt} + \epsilon_{ijt}, \tag{2}$$

Individuals observe g^e_{ijt} when making decisions, whereas the researcher does not. The expected utility of the outside good is normalized to $u^e_{i0t} = u_{i0t} = \epsilon_{i0t}$. Individuals choose the option that maximizes expected utility:

$$d_{ijt} \equiv \mathbb{1}\left\{u^e_{ijt} \geq \max_{j' \in \mathcal{J}} u^e_{ij't}\right\},$$

with $d_{ijt} = 1$ if i chooses option j in market t and $d_{ijt} = 0$ otherwise. We formalize these modeling assumptions as follows:

Assumption 1 (Choice under uncertainty)**.** *At the time of choice, individuals are uncertain only about g_{ijt} and form rational expectations conditional on $\mathcal{W}_{it}$. They choose the option that maximizes expected utility given in* (2)*. Taste shocks ϵ_{ijt} are i.i.d. across j conditional on (i, t) and are independent of $(X_{ijt}, \mathcal{W}_{it})$.*

The set $\mathcal{W}_{it}$ consists of information variables which we define as:

Definition 1. *A candidate variable w is:*

(a) ***Predictive*** *if $\mathbb{E}[g \mid w] \neq \mathbb{E}[g]$;*

(b) ***Used*** *if $w \in \mathcal{W}_{it}$, so that g^e_{ijt} depends on w.*

An information variable is a candidate variable that is both predictive and used.

In practice, information variables may include lagged realizations of the uncertain attribute, expert forecasts, or publicly available statistics related to g_{ijt}. As in revealed-preference settings such as CHN05, we do not distinguish between variables that are observed but not used by agents and variables the individual does not observe; both are treated as excluded from $\mathcal{W}_{it}$.

We formalize the prediction information set as

$$\mathcal{W}_{it} = \{w_{m,ijt} : m \in \mathcal{M}_i,\ j \in \mathcal{J}\},$$

where $\mathcal{M}_i$ denotes the index set of information variables used by individual i, and each $w_{m,ijt}$ may vary across options.[3]

[3]For simplicity, we assume that $\mathcal{M}_i$ is stable across markets. The framework can be extended to allow for market-specific information sets.

Remark 1. *Our framework allows observed attributes X_{ijt} to enter $\mathcal{W}_{it}$ and thus be used in forming predictions about g_{ijt}. For simplicity, we abstract from this going forward and assume that agents do not use X_{ijt} to predict g_{ijt}. This restriction allows us to distinguish between variables that shift utility directly and variables that operate exclusively through agents' information sets.*

Information heterogeneity arises in two ways. First, individuals may rely on different subsets of information variables, so that $\mathcal{M}_i \neq \mathcal{M}_{i'}$. Second, even when using the same information variables, realizations may differ across individuals. This structure follows DM18 in allowing both heterogeneity in information sets and heterogeneity in realizations. In contrast, CHN05 allow heterogeneous realizations but impose a common information set.

Assumption 2 (Information heterogeneity)**.** *The index set $\mathcal{M}_i$ is individual-specific and may not be common across individuals. Realizations of the information variables may also differ across individuals.*

The researcher observes a random sample of T markets. In each market t, the researcher observes either individual choices $d_t = \{d_{ijt}\}_{i\in\mathcal{N}_t, j\in\mathcal{J}}$ or aggregate market shares $\{s_{jt}\}_{j\in\mathcal{J}}$. Let $X_t = \{X_{ijt}\}_{i\in\mathcal{N}_t, j\in\mathcal{J}}$ denote the observed attributes and let $g_t = \{g_{ijt}\}_{i\in\mathcal{N}_t, j\in\mathcal{J}}$ denote the realizations of the uncertain attribute. Based on the data $\{d_t \text{ or } s_t, X_t, g_t\}$, the researcher's objective is to identify the preference parameters θ and the distribution of information sets $\mathcal{M}_i$ in the population. We close this section with two illustrative examples that serve as our empirical applications.

Example 1 (Exporting Firms)**.** The researcher observes individual-level data $\{d_t, X_t, g_t\}_{t\in\mathcal{T}}$. This corresponds to the setting in DM18, where the researcher observes the export decision d_{ijt} of each firm i to destination country j in year-sector t.

The utility function u_{ijt} in Equation (1) corresponds to exporter i's profit from exporting to j, which depends on observed attributes $X_{ijt} = X_{jt}$ such as bilateral distance. Profits also depend on export revenue g_{ijt}, which is uncertain at the time of the export decision. Each exporter therefore forms a prediction $g^e_{ijt} = \mathbb{E}[g_{ijt} \mid \mathcal{W}_{it}]$ based on its information set.

Assuming that the idiosyncratic error term ϵ_{ijt} follows a type-I extreme value distribution, the conditional choice probability given $\mathcal{W}_{it}$ from integrating over ϵ is

$$\Pr(d_{ijt} = 1 \mid X_t, \mathcal{W}_{it}; \theta) = \frac{\exp\left(X'_{jt}\beta + \gamma g^e_{ijt}\right)}{1 + \sum_{j'=1}^{J} \exp\left(X'_{j't}\beta + \gamma g^e_{ij't}\right)}.$$

Example 2 (Automobile Purchase)**.** The researcher observes market-level data $\{s_t, X_t, g_t\}_{t\in\mathcal{T}}$, as in Berry (1994) and Berry et al. (1995). For example, Grigolon et al. (2018) use market-level data to estimate consumers' valuation of fuel costs when purchasing automobiles.

The utility function u_{ijt} in Equation (1) corresponds to consumer i's utility from purchasing vehicle j in year–country t. Utility depends on observed attributes X_{jt} such as the vehicle price p_{jt}, and on fuel costs g_{jt} that are uncertain at the time of purchase. Each consumer forms a prediction $g^e_{ijt} = \mathbb{E}[g_{jt} \mid \mathcal{W}_{it}]$ based on her information set.

Assuming that ϵ_{ijt} follows a type-I extreme value distribution, the market share of product j in market t integrates the logit choice probability over the cross-sectional distribution of consumers' predictions. Let $g^e = (g^e_{1t}, \ldots, g^e_{Jt})$ denote the vector of predicted fuel costs across products in market t, and let $F_t(g^e)$ denote the distribution of this prediction vector across consumers. The predicted market share is

$$s_{jt}(X_t;\theta) = \int \frac{\exp\left(X'_{jt}\beta + \gamma g^e_{ijt}\right)}{1 + \sum_{j'=1}^{J} \exp\left(X'_{j't}\beta + \gamma g^e_{ij't}\right)} \, dF_t(g^e_{ijt}).$$

The researcher does not observe consumers' predictions g^e_{ijt} but observes realized fuel costs g_{jt}, constructed from mileage, fuel prices, and fuel economy.

3 Identification Challenges

This section explains why unobserved heterogeneity in agents' information sets creates fundamental identification challenges and why standard instrumental variable (IV) strategies generally fail in this environment.

Recall that agents' choices depend on expected attributes $g^e_{ijt} \equiv \mathbb{E}[g_{ijt} \mid \mathcal{W}_{it}]$, where $\mathcal{W}_{it}$ denotes the individual-specific information set used to form predictions. The researcher observes realized attributes g_{ijt} but not $\mathcal{W}_{it}$, nor the mapping from $\mathcal{W}_{it}$ to predictions g^e_{ijt}. When information sets are heterogeneous across agents as in Assumption 2, individuals facing identical realizations (X_{ijt}, g_{ijt}) may form different predictions and hence make different choices.

To illustrate, consider two individuals i and i' who face the same observed attributes and realizations of g_{ijt}, i.e. $X_{ijt} = X_{i'jt}$ and $g_{ijt} = g_{i'jt}$. However, individual i's information set contains information variable $w_{1,ijt}$ that is positively associated with g_{ijt}, while individual i' uses $w_{2,i'jt}$ which is negatively associated with $g_{i'jt}$. Then the two individuals may form

different predictions despite facing identical observed (X, g).

A common response is to treat the realized attribute g_{ijt} as endogenous and consider an instrumental-variables (IV) strategy. In this environment, however, agents' choices depend on their predictions g^e_{ijt}, rather than on the realization g_{ijt} itself. An IV approach therefore requires a variable that shifts agents' predictions while being excluded from all unobserved determinants of utility. Because predictions are formed conditional on the individual-specific information set $\mathcal{W}_{it}$, this exclusion restriction requires that the instrument be independent of every component of information that is unobserved by the researcher. When information sets are heterogeneous across individuals and their composition is unknown, this requirement becomes extremely restrictive.[4]

A common "plug-in" approach replaces the unobserved prediction g^e_{ijt} in (2) with the realized attribute g_{ijt}, effectively imposing perfect foresight. When agents are not perfectly informed, however, the unobserved prediction error $e_{ijt} \equiv g_{ijt} - g^e_{ijt}$ is non-zero, and variation in g_{ijt} reflects both variation in agents' predictions and variation in prediction errors. As a result, estimates that treat g_{ijt} as the relevant regressor confound preference sensitivity γ with the way predictions and prediction errors co-move with g_{ijt}.

More generally, realized outcomes g_{ijt} are a noisy proxy for the decision-relevant g^e_{ijt}, and the discrepancy e_{ijt} need not behave like classical measurement error. Because the covariance between e_{ijt} and g_{ijt} depends on agents' prediction information sets, replacing g^e_{ijt} with g_{ijt} can generate substantial bias of either sign. Section 6 and our empirical applications illustrate the magnitude of this bias.

Because choices depend on predictions rather than realizations, market shares alone do not identify the cross-sectional distribution of predictions g^e_{ijt} when the information structure is unobserved. The key issue is that γ governs choices through predictions, and the dispersion of these predictions across agents is unrestricted in the absence of additional structure. Bounding this dispersion yields partial identification of γ, whereas point identification requires stronger restrictions on how the distribution of predictions varies across individuals; these conditions are discussed in Appendix A.1. These insights motivate the identification strategies developed in the next two sections.

[4] Brown (2019); Brown et al. (2023) provide examples of an IV-style identification strategy in informational settings: their treatment z generates exogenous variation in the signal precision, which in turn shifts predictions g^e in a known way.

4 Point Identification with Finite Mixture Model

This section presents a semi-parametric approach that leverages restrictions on the data-generating process of the uncertain attribute g to address unobserved and heterogeneous predictions. We propose a two-step finite-mixture procedure. First, we construct conditional expectations of the uncertain attribute under alternative combinations of information variables. Each combination corresponds to a candidate information set and therefore defines a potential information type. Second, these types imply different predicted values of the uncertain attribute that enter the utility function, and we estimate the population shares of these latent types using a finite-mixture model.

4.1 Finite Mixture Model

To address the identification challenge posed by unobserved prediction information sets and information heterogeneity, we impose additional structure on how the uncertain attribute g_{ijt} relates to the information variables in $\mathcal{W}_{it}$. This structure restricts the support of unobserved predictions g^e_{ijt} and, in turn, disciplines the mapping from primitives to observed choice behavior.

Suppose the researcher observes a fixed list of candidate information variables $K^{\text{obs}} = \{k_1, \ldots, k_{|K^{\text{obs}}|}\}$ that are statistically related to the uncertain attribute g_{ijt}. For each product j, individual i, and market t, let

$$K^{\text{obs}}_{ijt} = (k_{1,ijt}, \ldots, k_{|K^{\text{obs}}|,ijt}) \in \mathbb{R}^{|K^{\text{obs}}|}$$

denote the realized vector of these variables. We refer to the elements of K^{obs} as *information candidates*. Each candidate variable $k \in K^{\text{obs}}$ satisfies the notion of *predictive information* in Definition 1(a): $\mathbb{E}[g_{ijt} \mid k] \neq \mathbb{E}[g_{ijt}]$.

We assume that agents form expectations about g_{ijt} using subsets of the predictive variables K^{obs}_{ijt} and allow agents to combine subsets of the observed candidates K^{obs}_{ijt} with latent components. In particular, individual-specific information sets $\mathcal{W}_{it}$ consist of subsets of K^{obs}_{ijt} and latent factors.

We specify the data-generating process of the uncertain attribute as

$$g_{ijt} = g\left(K^{\text{obs}}_{ijt}; \tau\right) + \nu_{ijt}, \tag{3}$$

where $g(\cdot; \tau)$ is a known function parameterized by τ and ν_{ijt} contains latent information

and an unobserved shock. Equation 3 describes the statistical process of g_{ijt} and does not depend on agents' predictions or choices. For tractability, our exposition is based on a linear specification for g_{ijt}, but the approach extends to more general functions $g(\cdot;\tau)$, which we explore in our empirical applications.

We further decompose the residual ν_{ijt} to allow for latent predictors of g_{ijt} that agents may use when forming predictions. Two common specifications are an additive fixed-effect structure and a linear factor structure (as in CHN05):

$$\nu_{ijt} = \nu_i + \nu_j + \nu_t + \eta_{ijt} \qquad \text{(additive fixed effects)}, \tag{4}$$

$$\nu_{ijt} = \lambda_i' F_{jt} + \eta_{ijt} \qquad \text{(linear factor model)}. \tag{5}$$

In (5), $\lambda_i \in \mathbb{R}^{|F|}$ denotes individual-specific loadings and $F_{jt} \in \mathbb{R}^{|F|}$ denotes option and market factors, with $|F|$ the number of factors. In either case, η_{ijt} is a residual satisfying mean-independence given the corresponding predictors (e.g., $\mathbb{E}[\eta_{ijt} \mid K_{ijt}^{\text{obs}}, \nu_i, \nu_j, \nu_t] = 0$ in (4) and $\mathbb{E}[\eta_{ijt} \mid K_{ijt}^{\text{obs}}, \lambda_i, F_{jt}] = 0$ in (5)).

For the remainder of the analysis, we work with an *augmented* list of information candidates that includes both observed and latent predictors:

$$K_{ijt} \equiv \left(K_{ijt}^{\text{obs}},\, K_{ijt}^{\text{unobs}}\right), \qquad K_{ijt}^{\text{unobs}} = \begin{cases} (\nu_i, \nu_j, \nu_t), & \text{under (4)}, \\ (\lambda_i, F_{jt}), & \text{under (5)}. \end{cases}$$

We focus on the linear factor model (5) in the rest of the paper; the arguments extend directly to the additive fixed-effect case.

Combining the specification choices above, we obtain the following data-generating process for the uncertain attribute:

$$g_{ijt} = K_{ijt}^{\text{obs}\prime}\tau + \lambda_i' F_{jt} + \eta_{ijt}. \tag{6}$$

We maintain the following assumption.

Assumption 3 (DGP for the uncertain attribute)**.** *The uncertain attribute g_{ijt} follows* (6)*. The researcher observes the list of observed information candidates K_{ijt}^{obs}. The remaining predictors of g_{ijt} are captured by the linear factor component $\lambda_i' F_{jt}$, and the residual satisfies $\mathbb{E}[\eta_{ijt} \mid K_{ijt}^{\text{obs}}, \lambda_i, F_{jt}] = 0$.*

Assumption 3 restricts the statistical relationship between the uncertain attribute and observable predictors while allowing agents to rely on additional latent signals summarized by the factor structure. This parallels the role of outcome processes in CHN05

and the maintained model for g in DM18, which enable the researcher to characterize how expectations vary across alternatives.[5]

To represent information heterogeneity across individuals, we consider all feasible prediction information sets that can be formed from the augmented list of candidate information variables. For each individual i and market t, let K_{it} denote the collection of candidate variables available to the individual, obtained from the augmented list $\{K_{ijt}\}_{j\in\mathcal{J}_t}$. Each element of K_{it} corresponds to one candidate predictor, which may itself vary across alternatives j.

Individual-specific prediction information sets $\mathcal{W}_{it}$ are assumed to consist of subsets of these candidate variables. We therefore consider all non-empty subsets of the $|K|$ candidate predictors as candidate prediction information sets used by different information types.

Formally, let $\mathcal{P}(K_{it})$ denote the collection of all non-empty subsets of the $|K|$ candidate variables, and index information types by $\kappa \in \mathcal{K} \equiv \{1, \ldots, |\mathcal{K}|\}$, where

$$|\mathcal{K}| = |\mathcal{P}(K_{it})| = \sum_{k=1}^{|K|} \binom{|K|}{k} = 2^{|K|} - 1 < \infty.$$

Each type κ is associated with a specific prediction information set $\mathcal{W}_{\kappa,it} \in \mathcal{P}(K_{it})$ and a corresponding prediction

$$g^e_{\kappa,ijt} \equiv \mathbb{E}[g_{ijt} \mid \mathcal{W}_{\kappa,it}].$$

This framework nests a full-information benchmark in which one type conditions on the complete set of candidate predictors K_{it}, yielding the prediction $\mathbb{E}[g_{ijt} \mid K_{it}]$ implied by the maintained data-generating process for g_{ijt}.

With the type-specific predictions $g^e_{\kappa,ijt}$, we model information heterogeneity using a finite-mixture specification. Let $\kappa(i) \in \mathcal{K}$ denote the *information type* of individual i, drawn from a type distribution

$$\phi = \left(\phi_1, \phi_2, \ldots, \phi_{|\mathcal{K}|}\right)' \in \mathbb{R}^{|\mathcal{K}|},$$

with probability mass function

$$\phi_\kappa \equiv \Pr\big(\kappa(i) = \kappa\big) \geq 0, \qquad \kappa \in \mathcal{K}, \qquad \sum_{\kappa=1}^{|\mathcal{K}|} \phi_\kappa = 1. \tag{7}$$

[5] In practice, the functional form of the mapping from observed predictors K^{obs}_{ijt} to the uncertain attribute g_{ijt} can be informed by the data using flexible statistical or machine learning methods. Our framework allows for such flexibility, as explored in the empirical applications.

We maintain the following assumption.

Assumption 4 (Finite-mixture of information types)**.** *Every agent is of an information type in $\mathcal{K}$, with $\kappa(i) \in \mathcal{K}$ drawn from the population distribution ϕ defined in* (7)*.*

Incorporating the finite-mixture structure into the demand system, both individual choice probabilities (with individual-level data) and market shares (with market-level data) can be written as mixtures of type-specific logit probabilities.[6] With individual-level data, the unconditional choice probability is

$$\Pr(d_{ijt} = 1) = \sum_{\kappa=1}^{|\mathcal{K}|} \phi_\kappa \frac{\exp\left(X'_{ijt}\beta + \gamma g^e_{\kappa jt}\right)}{1 + \sum_{j'=1}^{J} \exp\left(X'_{ij't}\beta + \gamma g^e_{\kappa j't}\right)}. \tag{8}$$

With market-level data, the predicted share of product j in market t is

$$s_{jt}(X_t, g^e_t; \Theta) = \sum_{\kappa=1}^{|\mathcal{K}|} \phi_\kappa \, s_{\kappa jt}(X_t, g^e_{\kappa t}; \theta), \tag{9}$$

where

$$s_{\kappa jt}(X_t, g^e_{\kappa t}; \theta) \equiv \frac{\exp\left(X'_{jt}\beta + \gamma g^e_{\kappa jt}\right)}{1 + \sum_{j'=1}^{J} \exp\left(X'_{j't}\beta + \gamma g^e_{\kappa j't}\right)}.$$

Here $g^e_{\kappa t} = (g^e_{\kappa 1t}, \ldots, g^e_{\kappa Jt})'$ denotes the vector of predicted attributes for type κ, and $\Theta \equiv (\theta', \phi')'$ collects preference parameters and mixture weights. Figure A1 illustrates the construction of type-specific prediction information sets and their mapping into type-specific predictions.

An important benefit of the framework is that it allows us to test the content of agents' prediction information sets. We can test for the presence of a given type κ by assessing whether the corresponding mixture weight ϕ_κ differs from zero. A positive and significant estimate of ϕ_κ implies that the information set $\mathcal{W}_{\kappa,it}$ is used by a non-negligible share of the population when forming predictions. This test can be implemented simultaneously for all types. Moreover, the estimated type shares $\{\phi_\kappa\}_{\kappa \in \mathcal{K}}$ may be of direct interest for counterfactuals and for evaluating information interventions.

Our approach extends CHN05. Their framework assumes a common information set

[6] Finite-mixture specifications have been used elsewhere to approximate the distribution of preference heterogeneity (e.g., Berry and Jia, 2010; Nevo et al., 2016; Bonhomme et al., 2022); here we instead use a finite mixture to approximate heterogeneity in predictions g^e_{ijt} induced by unobserved and heterogeneous information.

across individuals while permitting unobserved information factors. We additionally allow the composition of prediction information sets to vary across individuals. Our finite-mixture specification also differs from the minimal-information assumption in DM18. Rather than assuming that the researcher knows a set of observed information variables used by all individuals, we specify the data-generating process of the uncertain attribute g_{ijt} up to estimable fixed effects or factors. This yields an augmented set of observed and latent predictors that may enter agents' predictions, allowing for heterogeneous prediction information sets. Given this structure, the estimation procedure jointly recovers the distribution of information types and the preference parameters from revealed-preference data.

Finally, relative to approaches that impose perfect foresight or adopt a single plug-in information set, our framework allows for a discrete set of type-specific predictions $\{g^e_{\kappa jt}\}_{\kappa \in \mathcal{K}}$. This accommodates heterogeneous information types and allows the data to discipline their prevalence by estimating the mixture weights ϕ from observed choice patterns. We close this section with an example that illustrates our approach.

Example 3 (Market-level Data)**.** Suppose the uncertain attribute is common across individuals, $g_{ijt} = g_{jt}$, and admits the decomposition $g_{jt} = k_{1,jt} + k_{2,jt} + \eta_{jt}$, where $k_{1,jt}$ is the observed information candidates and $k_{2,jt}$ is the (estimated) latent factor. The pool of candidates is therefore $K_{ijt} = \{k_{1,jt}, k_{2,jt}\}$, so $|\mathcal{K}| = 2^2 - 1 = 3$ information types.

Let the three types be $\kappa \in \{A, B, C\}$, with type-specific information sets $\mathcal{W}_{A,jt} = \{k_{1,jt}\}$, $\mathcal{W}_{B,jt} = \{k_{2,jt}\}$, and $\mathcal{W}_{C,jt} = \{k_{1,jt}, k_{2,jt}\}$. Their type-specific predictions are

$$g^e_{\kappa,jt} \equiv \mathbb{E}[g_{jt} \mid \mathcal{W}_{\kappa,jt}],$$

which imply

$$g^e_{A,jt} = k_{1,jt} + \mathbb{E}[k_{2,jt} \mid k_{1,jt}], \quad g^e_{B,jt} = k_{2,jt} + \mathbb{E}[k_{1,jt} \mid k_{2,jt}], \quad g^e_{C,jt} = k_{1,jt} + k_{2,jt}.$$

Type C corresponds to the full-information benchmark. The market share in (9) becomes

$$s_{jt}(X_t, g^e_t; \Theta) = \sum_{\kappa \in \{A,B,C\}} \phi_\kappa \, \frac{\exp\left(X'_{jt}\beta + \gamma g^e_{\kappa,jt}\right)}{1 + \sum_{j'=1}^{J} \exp\left(X'_{j't}\beta + \gamma g^e_{\kappa,j't}\right)}.$$

4.2 Identification

We focus on market-level data $\{s_t, X_t, g_t^e\}_{t\in\mathcal{T}}$. Let $s_t = (s_{0t}, s_{1t}, \ldots, s_{Jt})'$ denote observed shares in market t. Let $s_{\kappa jt}(\theta)$ denote the type-specific choice probability implied by the structural parameter θ. Stack these probabilities into the $(J+1) \times |\mathcal{K}|$ matrix $A_t(\theta)$ with entries $(A_t(\theta))_{j\kappa} = s_{\kappa jt}(\theta)$. The mixture structure implies the market share mapping

$$s_t = A_t(\theta)\phi.$$

Recovering $A_t(\theta)$ requires identifying the latent fixed effects/factors in the DGP for g_{ijt} (up to normalization), which we assume is feasible under standard factor-model conditions.

Assumption 5. *The type-specific choice probability matrix $A_t(\theta) \in \mathbb{R}^{(J+1)\times|\mathcal{K}|}$ has full column rank for at least one market t.*

Assumption 6. *The Jacobian matrix $B_t(\theta_0, \phi_0) = \left(\frac{\partial s_t(\theta_0,\phi_0)}{\partial\theta},\ A_t(\theta_0)\right) \in \mathbb{R}^{(J+1)\times(|\theta|+|\mathcal{K}|)}$ has full column rank for at least one market t.*

Theorem 1. *Consider the model defined by Assumptions 1–4. Under Assumption 5, the mixture weights ϕ_0 are uniquely identified given θ.*

If, in addition, Assumption 6 holds, then (θ_0, ϕ_0) is locally identified.

Proof. Under modeling Assumptions 1-4, the type-specific predictions $g^e_{\kappa,jt}$ are determined by the information structure. Hence, for any candidate θ, the type-specific choice probabilities $s_{\kappa jt}$ are known functions and $s_t(\theta, \phi) = A_t(\theta)\phi$.

First, under Assumption 5, $A_t(\theta_0)$ has full column rank. Hence the linear system $s_t = A_t(\theta_0)\phi$ admits a unique solution for ϕ, implying identification of ϕ_0 given θ_0.

Second, define

$$F(\theta, \phi) \equiv s_t(\theta, \phi) - s_t = 0, \qquad s_t(\theta, \phi) = A_t(\theta)\phi.$$

The Jacobian of F with respect to (θ, ϕ) at (θ_0, ϕ_0) is

$$B_t(\theta_0, \phi_0) = \left[\frac{\partial s_t}{\partial\theta}(\theta_0, \phi_0), \frac{\partial s_t}{\partial\phi}(\theta_0, \phi_0)\right] = \left[\frac{\partial s_t}{\partial\theta}(\theta_0, \phi_0), A_t(\theta_0)\right].$$

By Assumption 6, this matrix has full column rank. The implicit function theorem then implies that the solution to $F(\theta, \phi) = 0$ is locally unique in a neighborhood of (θ_0, ϕ_0), establishing local identification of (θ_0, ϕ_0). ■

We present Assumptions 5 and 6 such that they rely solely on within-market variation. In particular, full column rank requires $J+1 \geq |\mathcal{K}|$ for $A_t(\theta)$ and $J+1 \geq |\theta| + |\mathcal{K}|$ for $B_t(\theta, \phi)$. The rank condition for $A_t(\theta)$ requires that type-specific choice probabilities $s_{\kappa jt}(\theta)$ differ across κ, which in turn requires that the induced predictions $g^e_{\kappa jt}$ are not collinear and that $\gamma \neq 0$. If a single market does not provide sufficient variation, identification can instead rely on cross-market variation. Let $\mathcal{A}(\theta)$ and $\mathcal{B}(\theta, \phi)$ denote the matrices obtained by stacking $A_t(\theta)$ and $B_t(\theta, \phi)$ across markets. The rank conditions in Assumptions 5 and 6 then apply to these stacked matrices (which is generally easier to satisfy).

Theorem 1 characterizes identification in a choice model with unobserved information heterogeneity and provides conditions for point identification. Because choices depend on expected attributes, heterogeneity in information sets induces cross-sectional dispersion in predictions. Absent restrictions on the distribution of these predictions, observed choice variation cannot disentangle taste heterogeneity from belief heterogeneity, and preference parameters are not point identified.[7] We show that point identification can be restored by imposing structure on the data-generating process of the uncertain attribute, which restricts the support of prediction heterogeneity and generates observable, type-specific regressors. This yields a finite-mixture representation with known support and permits explicit rank conditions for identification.

Our semi-parametric finite-mixture approach relies on Assumption 3, which requires the correct specification of the data-generating process of the uncertain attribute g. In Section 5, we relax this assumption and allow the researcher to observe only a subset of the information variables. With weaker restrictions on the distribution of predictions, we obtain set identification of preference parameters rather than point identification. Finally, having established identification of (θ, ϕ) under the baseline finite-mixture specification, we now consider several extensions that relax auxiliary assumptions and broaden the scope of the model. Specifically, we allow for (i) market-specific information distributions, (ii) diagnostic tests of the information structure, and (iii) endogenous product characteristics as well as preference heterogeneity.

4.3 Extensions

Extension 1 (Market-specific information types and shares)**.** Identification of ϕ conditional on θ does not require cross-market variation. The mixture weights may therefore vary by

[7] See Barseghyan et al. (2021); Molinari (2020); Agarwal and Somaini (2022) for related approaches that separate preferences and beliefs in other environments; unlike our framework, these papers focus on settings with unobserved choice sets rather than unobserved information over attributes.

market, ϕ_t, provided that the rank condition on $A_t(\theta)$ holds within each market. Likewise, the number of information types $|\mathcal{K}|$ may differ across markets if the set of available information variables changes across markets. This extension permits the information distribution to vary with the market environment without altering the identification argument.

Extension 2 (Testing for misspecification of information)**.** The framework permits a diagnostic test of Assumption 3. Estimating the DGP for g_{ijt} yields residuals $\hat{\eta}_{ijt}$. Under correct specification, η_{ijt} contains no predictive content for g_{ijt} beyond the information candidates in K_{ijt} and therefore should not affect choices once K_{ijt} is accounted for.

To implement this test, the researcher can augment the mixture with an additional type κ_η whose prediction information set consists only of η_{ijt}. Under correct specification, the associated mixture weight ϕ_{κ_η} should equal zero. This mirrors the logic in CHN05: once all choice-relevant information is included, remaining variation in the uncertain attribute should be orthogonal to choices.

One limitation is that a positive weight on the residual type is insufficient to indicate misspecification. If some agents observe g_{ijt} (or additional signals correlated with η_{ijt}), then η_{ijt} becomes choice-relevant even under correct specification of the researcher's information candidates, and the mixture may assign positive mass to κ_η to capture such full-information (or extra-information) types. In that case, $\phi_{\kappa_\eta} > 0$ confounds omitted information in the researcher's model with genuinely richer information available to some agents.

Extension 3 (Preference heterogeneity, endogeneity, and choice-specific unobservables)**.** We extend the framework to allow for unobserved preference heterogeneity, endogenous characteristics, and choice-specific unobservables. Following Berry and Haile (2014), we work with market-level data. Let $x_{jt}^{(1)}$ denote exogenous characteristics, p_{jt} an endogenous attribute (e.g. price), ξ_{jt} an unobserved demand shock, and z_{jt} instruments excluded from ξ_{jt}. Let $\tilde{x}_{jt}^{(2)}$ denote characteristics with heterogeneous tastes. We also allow for individual-specific sensitivity γ_i to the predicted attribute $g^e_{\kappa(i),jt}$.

Expected utility becomes

$$u^e_{ijt} = x_{jt}^{(1)\prime}\beta - \alpha p_{jt} + \gamma_i g^e_{\kappa(i),jt} + \tilde{x}_{jt}^{(2)\prime}\beta_i^{(2)} + \xi_{jt} + \epsilon_{ijt} \tag{10}$$

The random coefficients $(\beta_i^{(2)}, \gamma_i)$ follow a distribution $\tilde{F}_{\beta^{(2)},\gamma}$. Importantly, the information structure restricts the prediction component to lie on a finite set of known directions determined by the information types. This transforms Assumption 1 into:

Assumption 1′ (Choice under uncertainty′)**.** *At the time of choice, individuals are uncertain only about g_{ijt} and form rational expectations conditional on $\mathcal{W}_{it}$. They choose the option that maximizes expected utility given in* (10)*. Taste shocks ϵ_{ijt} are i.i.d. across j conditional on (i,t) and are independent of $(x_{jt}^{(1)}, \tilde{x}_{jt}^{(2)}, p_{jt}, \mathcal{W}_{it})$.*

Before we establish identification, we first show that our information model can be mapped into a standard random utility framework, which allows us to leverage results in Berry and Haile (2014) and Wang (2023). A key feature of our framework is that information heterogeneity generates type-specific regressors $g^e_{\kappa(i),jt}$ that enter utility with a common coefficient structure. This yields a random-coefficients representation with known finite support induced by the information types. This is formalized as follows:

Proposition 1. *Under Assumptions 1′–4, the model* (10) *can be written as a standard random utility model with utility*

$$u^e_{ijt} = x_{jt}^{(1)\prime}\beta - \alpha p_{jt} + x_{jt}^{(2)\prime}\beta_i + \xi_{jt} + \epsilon_{ijt},$$

where $x_{jt}^{(2)}$ stacks the taste-shifter characteristics $\tilde{x}_{jt}^{(2)}$ and the vector of type-specific prediction regressors $\tilde{g}^e_{jt}$, and β_i is a vector of random coefficients distributed according to F.

The coefficient on the prediction component admits the representation

$$\tilde{\gamma}_i = \gamma_i b_{\kappa(i)},$$

where γ_i is an individual-specific scale parameter and $\kappa(i) \in \mathcal{K}$ denotes the information type drawn from the distribution ϕ. The vectors $\{b_\kappa\}_{\kappa\in\mathcal{K}}$ are known and determine the finite set of directions induced by the information structure.

Hence information heterogeneity enters the random-coefficients representation through a finite set of directions $\{b_\kappa\}_{\kappa\in\mathcal{K}}$, corresponding to the information types, while the magnitude γ_i remains unrestricted.

Proof. See Appendix A.2 ■

Assumption 7 (Exogeneity of random coefficients and information types)**.** *The pair $(\beta_i^{(2)}, \gamma_i)$ is distributed according to $\tilde{F}_{\beta^{(2)},\gamma}$ and is independent of $(x_{jt}^{(1)}, \tilde{x}_{jt}^{(2)}, p_{jt}, \xi_{jt})_{j\in\mathcal{J}_t\setminus\{0\}}$ and $(\epsilon_{ijt})_{j\in\mathcal{J}_t}$. Moreover, $(\beta_i^{(2)}, \gamma_i)$ is independent of $\kappa(i)$.*

Assumption 7 imposes standard exogeneity restrictions on preference heterogeneity and information types. The distribution $\tilde{F}_{\beta^{(2)},\gamma}$ governing $(\beta_i^{(2)}, \gamma_i)$ is independent of

product characteristics, prices, and unobserved demand shocks. Likewise, the distribution of information types ϕ is independent of market-level primitives. The assumption that $(\beta_i^{(2)}, \gamma_i)$ is independent of $\kappa(i)$ rules out endogenous sorting of consumers with particular tastes into specific information types and ensures that information heterogeneity affects choices only through the prediction component of utility. These restrictions follow standard random-coefficients frameworks such as Wang (2023). Similar independence assumptions appear in the literature on limited consideration (e.g., Abaluck and Adams-Prassl, 2021), where relaxing the independence between attention and utility shocks generally requires additional instruments.[8,9]

Theorem 2. *Under Assumptions 1′–4 and 7, Assumptions 1–4 of Berry and Haile (2014), and Condition 2 of Wang (2023) applied to model* (10)*, the primitives* $(\beta_0, \alpha_0, \phi_0, \tilde{F}_{\beta^{(2)},\gamma})$ *are identified.*

Proof. See Appendix A.3 ■

The key condition enabling this result is that the information structure implies a random-coefficients representation in which the prediction coefficient lies on a finite set of known directions determined by the information types.

Under Proposition 1, Assumptions 1–4 of Berry and Haile (2014) imply nonparametric identification of the demand system. The key restriction is instrument exogeneity (Assumption 3 in Berry and Haile, 2014), which in our setting requires that the predicted attributes $g^e_{\kappa(i),jt}$ be conditionally mean independent of the demand shocks ξ_{jt}. This holds if the information variables $\mathcal{W}_{jt}$ and the information types $\kappa(i)$ are independent of ξ_{jt} conditional on $(z_t, x_t^{(1)})$, so that they provide valid instruments.

Condition 2 of Wang (2023) requires a rank condition on the random-coefficient regressors. In our setting this requires that the columns of the prediction vector $\tilde{g}^e_t$ are not perfectly collinear across information types (analogous to Assumption 5 in our benchmark model). This condition identifies the distribution of the random coefficient associated with the prediction component. Because our framework decomposes this coefficient into a magnitude γ_i and a direction determined by the information type $\kappa(i)$, Assumption 7 allows us to separately identify the taste distribution $\tilde{F}_{\beta^{(2)},\gamma}$ and the mixture weights ϕ governing

[8] "[...] both models impose that the unobservable determinants of attention and utility are uncorrelated. This restriction can be relaxed but not without additional instruments" (p. 1624, Abaluck and Adams-Prassl, 2021).

[9] We leave the identification with endogenous formation of information types for future work. Barseghyan et al. (2021) explicitly allow unrestricted dependence (conditional on observables) between unobserved choice sets and preferences, and obtain partial identification via moment inequalities. Brown and Jeon (2024) also allow for endogenous sorting into information types with a full structural model of rational inattention.

information types. Thus information heterogeneity generates additional variation in market shares across types, which identifies ϕ and sharpens identification of the full parameter vector relative to models with unrestricted random coefficients.

4.4 Estimation

Estimation with Individual-level Data. Given individual-level data, we estimate the parameters of interest by maximum likelihood. Let d_{ijt} denote an indicator equal to one if individual i chooses product j in market t, with $\sum_{j=0}^{J} d_{ijt} = 1$. The likelihood function is

$$\mathcal{L}\Theta|d) = \prod_{i,t} \prod_{j=0}^{J} \left[\Pr(d_{ijt} = 1 \mid \Theta)\right]^{d_{ijt}}. \tag{11}$$

The choice probabilities $\Pr(d_{ijt} = 1 \mid \Theta)$ correspond to the mixture logit probabilities defined in Equation (8). When the choice set contains only a single product ($J = 1$), the multinomial likelihood reduces to the binary-choice likelihood used in DM18.

Estimation with Market-level Data. With market-level data, we extend the GMM approach of Berry et al. (1995). Let $\xi_{jt}(\Theta)$ denote the structural error (mean utility residual) implied by the model at parameter vector $\Theta = (\theta', \phi')'$. We impose mean independence of $\xi_{jt}(\Theta)$ with respect to a vector of instruments $Z_{jt} \in \mathbb{R}^{|Z|}$:

$$\mathbb{E}\left[\xi_{jt}(\Theta_0) \mid Z_{jt}\right] = 0,$$

where $|Z|$ denotes the number of instruments. The instrument vector Z_{jt} stacks (i) information shifters used to construct type-specific predictions, (ii) exogenous observed attributes, and (iii) additional excluded instruments for endogenous components of X_{jt} (e.g., cost shifters for prices).

The estimator chooses Θ to minimize a quadratic form in the stacked sample moments. Let $\xi(\Theta)$ and Z denote the vectors/matrices stacked over (j, t), and let W be a (consistent) estimate of the optimal weighting matrix. We solve

$$\begin{aligned} \min_{\Theta} \quad & \left(\frac{1}{N} Z'\xi(\Theta)\right)' W \left(\frac{1}{N} Z'\xi(\Theta)\right) \\ \text{s.t.} \quad & \phi_\kappa \geq 0 \ \ \forall \kappa \in \mathcal{K}, \qquad \sum_{\kappa \in \mathcal{K}} \phi_\kappa = 1, \end{aligned}$$

where N is the number of (j, t) observations used in the moments.

In principle, the information shifters $\{k_{mjt}\}$ are potential instruments. In practice, however, they often perform poorly because they enter the moments only through the non-linear mapping from information to predicted attributes and then to shares. Moreover, the researcher may include candidate information variables (or types) that are irrelevant because they are not used by any agents, further weakening standard IV variation.

To address both issues, we use Chamberlain's approximation to optimal instruments (Chamberlain, 1987), which is proportional to the conditional expectation of the score/Jacobian, $\mathbb{E}\left[\partial\xi_{jt}(\Theta)/\partial\Theta' \mid Z_{jt}\right]$. These instruments (i) apply a non-linear operator to the baseline instrument set Z_{jt}, capturing the non-linear way information affects shares, and (ii) target the parameters of interest directly (notably ϕ and γ). If a candidate type κ is irrelevant, then $\partial\xi_{jt}(\Theta)/\partial\phi_\kappa$ is approximately zero, so the corresponding optimal instrument contributes negligibly, effectively down-weighting non-existing types.

Following Reynaert and Verboven (2014), we implement a two-stage approximation. In the first stage we estimate the model using baseline instruments Z_{jt} (including information shifters and standard demand instruments). We then construct empirical Chamberlain instruments by evaluating $\mathbb{E}\left[\partial\xi_{jt}(\Theta)/\partial(\phi', \gamma)' \mid Z_{jt}\right]$ at the first-stage estimates. In the second stage, we either (i) run a standard two-step GMM that treats the constructed Chamberlain instruments as fixed and uses an updated optimal weighting matrix, or (ii) use a continuously-updated GMM (CUE) estimator that re-computes the Chamberlain instruments over the minimization. In our applications, the CUE approach is often more stable in mixture models because it keeps the instruments aligned with the current parameter values along the optimization path.

5 Moment Inequalities from Extremal Information

We now develop a partial-identification approach for inference on preference parameters that relaxes Assumption 3 and is feasible in the presence of many types. The key idea is simple: even if we do not observe agents' predictions g^e_{jt}, revealed preference implies bounds on preferences once we can isolate states in which an *observed* prediction rule delivers extremal prediction errors. In those extreme states, the observed rule provides a worst-case (or best-case) approximation to the unobserved prediction entering utility, generating moment inequalities that are valid under heterogeneous and unobserved prediction information sets. This approach is implementable with market-level data, whereas existing moment-inequality methods such as DM18 are developed for individual-level data.

Formally, we show that partial knowledge of the cross-sectional distribution of prediction

errors–in particular, its behavior in the tails conditional on observable information shifters–is sufficient to set identify θ. Operationally, the researcher needs (i) to observe the prediction information set of at least one information type that is present in the data, and (ii) to construct a selection rule that picks observations in which this observed type attains the maximal or minimal prediction error among all types. These extremal events allow us to replace unknown bounds driven by unobserved information with observable bounds driven by the selected type, yielding inequalities that are informative about θ.

5.1 Identification

We start by defining conditional moment inequalities given a vector of conditioning variables and instruments. Let Z_{jt} denote the vector observed by the researcher used for conditioning. It may include (i) shifters used to form observable predictions $g^e_{\tilde{\kappa}jt}$ for an observed information type $\tilde{\kappa}$, and (ii) standard demand instruments used to address endogeneity in X_{jt} through the unobserved demand shock ξ_{jt}. We consider the conditional inequalities

$$\mathbf{m}_{\max}(Z_{jt};\theta) \geq 0, \qquad \mathbf{m}_{\min}(Z_{jt};\theta) \geq 0, \tag{12}$$

where

$$\begin{aligned} \mathbf{m}_{\max}(Z_{jt};\theta) &\equiv \mathbb{E}\left[-\log\left(\frac{s_{jt}}{s_{0t}}\right) + X'_{jt}\beta + \gamma g_{jt} + \xi_{jt} + \max_{\kappa\in\mathcal{K}_t}\{-\gamma e_{\kappa jt}\} \;\middle|\; Z_{jt} \right], \\ \mathbf{m}_{\min}(Z_{jt};\theta) &\equiv \mathbb{E}\left[\log\left(\frac{s_{jt}}{s_{0t}}\right) - X'_{jt}\beta - \gamma g_{jt} - \xi_{jt} - \min_{\kappa\in\mathcal{K}_t}\{-\gamma e_{\kappa jt}\} \;\middle|\; Z_{jt} \right]. \end{aligned} \tag{13}$$

Let Θ denote the parameter space for $\theta = (\beta', \gamma)'$. Define the identified set

$$\Theta_0 \equiv \Big\{ \theta \in \Theta : \mathbf{m}_{\max}(Z_{jt};\theta) \geq 0 \text{ and } \mathbf{m}_{\min}(Z_{jt};\theta) \geq 0 \text{ for almost every } Z_{jt} \Big\}.$$

Theorem 3. *Assume a finite-mixture logit structure holds. Then the population conditional inequalities in* (13) *hold. In particular, the true parameter* $\theta_0 \in \Theta_0$.

Theorem 3 states that the inequalities are satisfied at the true parameter value. The proof is in Appendix A.4. The intuition is clearest from the equivalent bounds on the observed log share ratio $\Delta_{jt} \equiv \log(s_{jt}/s_{0t})$:

$$\gamma g_{jt} + X'_{jt}\beta + \xi_{jt} + \min_{\kappa\in\mathcal{K}_t}\{-\gamma e_{\kappa jt}\} \;\leq\; \Delta_{jt} \;\leq\; \gamma g_{jt} + X'_{jt}\beta + \xi_{jt} + \max_{\kappa\in\mathcal{K}_t}\{-\gamma e_{\kappa jt}\}. \tag{14}$$

In words, for fixed (X_{jt}, g_{jt}), the log-odds must lie between the smallest and largest utility

indices induced by feasible prediction errors across information types.[10]

The two bounds in (14) are complementary. Consider the two non-negative inequalities constructed from (14). Holding fixed all parameters except γ and taking $\gamma < 0$ (wlog), the non-negative inequality involving the right-hand side is decreasing in γ and hence delivers an upper bound on γ, while the one involving the left-hand side is increasing in γ and delivers a lower bound.

Inequalities (14) are not directly implementable because they involve the extremal prediction errors $\max_\kappa e_{\kappa jt}$ and $\min_\kappa e_{\kappa jt}$. We therefore work with *extremal-error events* that allow the researcher to replace these extrema by the prediction error of an observed information type.[11]

Definition 2 (Extremal-error indices)**.** *Let $e_{\kappa jt} \equiv g_{jt} - g^e_{\kappa jt}$ denote the prediction error of type κ for option j in market t. Define*

$$\kappa_{\max}(j,t) \in \arg\max_{\kappa \in \mathcal{K}_t} e_{\kappa jt}, \qquad \kappa_{\min}(j,t) \in \arg\min_{\kappa \in \mathcal{K}_t} e_{\kappa jt},$$

These indices locate the types that attain the maximal/minimal prediction error.

Rather than attempting to recover $\kappa_{\max}(j,t)$ and $\kappa_{\min}(j,t)$ for each observation, we assume the researcher observes the prediction information set for at least one information type $\tilde{\kappa}$ that exists in the data, and can construct selection indicators that isolate observations where the observed $\tilde{\kappa}$ attains the maximal or minimal error.

Let Z^P_{jt} denote standard demand instruments for endogenous components of X_{jt} (e.g., cost shifters). We collect conditioning variables in

$$Z_{jt} \equiv \left(Z^P_{jt},\, X_{jt},\, \mathcal{W}_{\tilde{\kappa} jt}\right).$$

Assumption 8 (Extremal-error selection)**.** *Assume:*

(i) The researcher observes $\mathcal{W}_{\tilde{\kappa} jt}$ for at least one information type $\tilde{\kappa}$ that exists in the data.

(ii) There exists an indicator $Z^S_{\max\, jt} \in \{0,1\}$ such that, on selected observations, $\tilde{\kappa}$ attains the maximal prediction error in the sense that

$$\mathbb{E}\left[e_{\tilde{\kappa} jt}\, Z^S_{\max\, jt} \mid Z_{jt}\right] = \mathbb{E}\left[\max_{\kappa \in \mathcal{K}_t} e_{\kappa jt}\, Z^S_{\max\, jt} \mid Z_{jt}\right].$$

[10]The inequalities are closely related to the linear conditional moment-inequality approach in Gandhi et al. (2023), who use market-level data to conduct robust inference in settings with zero market shares.

[11]Extremality refers to the realized prediction error $e_{\kappa jt}$ and does not imply that the corresponding type has more or less accurate information in general.

(iii) There exists an indicator $Z^S_{\min\, jt} \in \{0,1\}$ such that, on selected observations, $\tilde{\kappa}$ attains the minimal prediction error in the sense that

$$\mathbb{E}\left[e_{\tilde{\kappa}jt}\, Z^S_{\min\, jt} \mid Z_{jt}\right] = \mathbb{E}\left[\min_{\kappa\in\mathcal{K}_t} e_{\kappa jt}\, Z^S_{\min\, jt} \mid Z_{jt}\right].$$

(iv) The unobserved demand shock is mean independent on the selected observations:

$$\mathbb{E}\left[\xi_{jt}\, Z^S_{\max\, jt} \mid Z_{jt}\right] = 0, \qquad \mathbb{E}\left[\xi_{jt}\, Z^S_{\min\, jt} \mid Z_{jt}\right] = 0.$$

A natural selection rule uses thresholds on the observed-type error:

$$Z^S_{\max\, jt} = \mathbb{1}\{e_{\tilde{\kappa}jt} \geq a_{\max}\}, \qquad Z^S_{\min\, jt} = \mathbb{1}\{e_{\tilde{\kappa}jt} \leq a_{\min}\},$$

where the thresholds can be chosen as extreme quantiles of the empirical distribution of $e_{\tilde{\kappa}jt}$. Tighter thresholds reduce the chance of selecting non-extremal observations (bias) but increase sampling variability by reducing the effective sample size (variance).

Applying these selection indicators to (13) yields implementable conditional moment inequalities that depend only on the observed prediction $g^e_{\tilde{\kappa}jt}$:

$$\begin{aligned}
\mathbf{m}^{\tilde{\kappa}}_{\max}(Z_{jt};\theta) &\equiv \mathbb{E}\left[\left(-\log\frac{s_{jt}}{s_{0t}} + \gamma\, g^e_{\tilde{\kappa}jt} + X'_{jt}\beta + \xi_{jt}\right)\cdot Z^S_{\max\, jt} \;\middle|\; Z_{jt}\right] \geq 0, \\
\mathbf{m}^{\tilde{\kappa}}_{\min}(Z_{jt};\theta) &\equiv \mathbb{E}\left[\left(\log\frac{s_{jt}}{s_{0t}} - \gamma\, g^e_{\tilde{\kappa}jt} - X'_{jt}\beta - \xi_{jt}\right)\cdot Z^S_{\min\, jt} \;\middle|\; Z_{jt}\right] \geq 0,
\end{aligned} \tag{15}$$

where $g^e_{\tilde{\kappa}jt} \equiv \mathbb{E}[g_{jt} \mid \mathcal{W}_{\tilde{\kappa}jt}]$ is the prediction implied by the observed information type $\tilde{\kappa}$.[12]

Example 2 (continued) (Automobile Purchase)**.** Consider the fuel-cost setting of Example 2 and Grigolon et al. (2018), where the uncertain attribute is realized fuel cost per mile $g_{jt} = e_{jt} f_t$ and consumers form predictions $g^e_{jt} = \mathbb{E}[g_{jt} \mid \mathcal{W}]$ from subsets of the observable predictors $\{e_{jt}, f_t\}$. Suppose there exists an information type $\tilde{\kappa}$ that predicts fuel costs using fuel economy only, $\mathcal{W}_{\tilde{\kappa}} = \{e_{jt}\}$. When fuel prices are sufficiently high, this type's omitted fuel-price information generates large prediction errors. The extremal-selection condition required for (15) is that, on a subset of such high-price observations, type $\tilde{\kappa}$ attains the maximal prediction error among all types.

[12]The inequalities are written assuming $\gamma \geq 0$. When $\gamma < 0$, the relevant extremal event reverses, so the roles of $Z^S_{\max\, jt}$ and $Z^S_{\min\, jt}$ are interchanged. The procedure therefore applies without loss of generality to either sign of γ.

5.2 Estimation

The estimation of the moment-inequality approach proceeds in three steps. First, we fix an observed information type indexed by $\tilde{\kappa} \in \mathcal{K}$ and specify its information set $\mathcal{W}_{\tilde{\kappa}t}$. Using this information set, we estimate the type-specific predictions $g^e_{\tilde{\kappa}jt} = \mathbb{E}[g_{jt} \mid \mathcal{W}_{\tilde{\kappa}t}]$ and the associated prediction errors $e_{\tilde{\kappa}jt} = g_{jt} - g^e_{\tilde{\kappa}jt}$. Second, we construct a selection indicator Z^S_{jt} that targets observations with extreme values of $e_{\tilde{\kappa}jt}$. Here the conditioning set is written at the market level because the signals used to form predictions are common across individuals within a market. Second, we form a selection indicator Z^S_{jt} that targets observations with extreme values of $e_{\tilde{\kappa}jt}$. The choice of Z^S_{jt} induces a bias–variance trade-off. A tighter selection criterion (e.g., more extreme percentiles) increases the likelihood that $\tilde{\kappa}$ coincides with the true extremal-error type on the selected observations, strengthening the plausibility of the extremal-selection restriction; however, it reduces the effective sample size and increases the sampling variability of the resulting moments. Conversely, a looser selection criterion retains more observations and lowers variance, but it increases the chance that some selected observations violate the extremal-selection restriction because $\tilde{\kappa}$ is not the extremal type, inducing bias in finite samples. We quantify this trade-off in the simulation study in Section 6. Third, we adapt the conditional moment-inequality inference framework of Andrews and Shi (2013) to construct confidence regions for the preference parameters based on the inequalities in (15).[13]

6 Simulations

We conduct simulations of both the finite mixture and moment-inequalities approach to illustrate their power. We focus on settings with market-level data.

6.1 Setup

Our simulation is based on the example described in Section 4.1 with three information types. We perform a Monte Carlo exercise with 1,000 datasets, each consisting of $T = 150$ markets and $J = 10$ products. The indirect utility is defined as

$$u^e_{ijt} = \beta_0 + \beta_x x_{jt} - \alpha p_{jt} + \gamma g^e_{jt} + \xi_{jt} + \epsilon_{ijt}.$$

[13] Andrews et al. (2023) and Cox and Shi (2023) provide alternative inference procedures for linear conditional moment inequalities, but Andrews and Shi (2013) is sufficiently powerful for our purposes and performs well in our simulations.

The product characteristic is uniformly distributed $x_{jt} \overset{\text{iid}}{\sim} U(1,2)$. We simulate the price $p_{jt} = 1 + z_{jt} + \nu_{jt}$ to be exogenous with a cost shifter $z_{jt} \overset{\text{iid}}{\sim} U(0,1)$ and a cost shock $\nu_{jt} \overset{\text{iid}}{\sim} U(-0.25, 0.25)$. The uncertain attribute is assumed to be constant across individuals $g_{ijt} = g_{jt}$ and equals the sum of two information variables and one random error $g_{jt} = k_{1jt} + k_{2jt} + \varepsilon^g_{jt}$. The three components are independent of each other. The first information variable is assumed to be observed by the researcher and follows a log-normal distribution $\log(k_{1jt}) \overset{\text{iid}}{\sim} N(0,1)$. The second information variable is unobserved by the researcher and follows a linear factor model $k_{2jt} = \lambda_j' F_t$ with $\lambda_j \overset{\text{iid}}{\sim} N(1,1)$ and an AR(1) process $F_t = 0.5F_{t-1} + \varepsilon^F_t$, $\varepsilon^F_t \overset{\text{iid}}{\sim} N(0,1)$. The random shock to the uncertain attribute follows a normal distribution $\varepsilon^g_{jt} \overset{\text{iid}}{\sim} N(0, 0.2^2)$. The demand shifters are uniformly distributed $\xi_{jt} \overset{\text{iid}}{\sim} U(-1,1)$. The idiosyncratic demand shocks ϵ_{ijt} follow a standard EVT1 distribution. The preference parameters are $\theta = (\beta_0, \beta_x, \alpha, \gamma)' = (1, 1, 1.5, -1.5)'$.

There are three information types in the simulations, indexed by $\kappa = A, B, C$, where the information set of each type is $\mathcal{W}_{At} = \{k_{1jt}\}$, $\mathcal{W}_{Bt} = \{k_{2jt}\}$, and $\mathcal{W}_{Ct} = \{k_{1jt}, k_{2jt}\}$. The proportion of each information type is $\phi_A = 0.15, \phi_B = 0.5, \phi_C = 0.35$. Here, the information type C corresponds to a full information type, and the information type B only uses the information variable that is unobserved by the researcher.

We also provide simulations where we introduce preference heterogeneity through random coefficients on the valuation of expected attributes γ. The random coefficient is specified as $\gamma_i = \gamma + \sigma_\gamma v_i$ where $\gamma = -1.5$ and $\sigma_\gamma = 0.5$. The distribution of γ_i is constructed by 100 Halton draws of v_i from the standard normal distribution and we have $\gamma_i \overset{\text{iid}}{\sim} N(-1.5, 0.5)$.

Finally, we simulate a third simulation with datasets that only contain two information types $\kappa = B, C$ with fractions $\phi_B = 0.4, \phi_C = 0.6$ to test if our method can recover consistent estimates when there are fewer information types in the data than specified. We test the method in the most challenging case where the DGP does not contain the information type A that uses only the observed information variable. To ensure that this setup generates a similar average market share of the outside option as the previous data, we adjust the distribution of the exogenous attribute $x_{jt} \overset{\text{iid}}{\sim} U(0,1)$.

6.2 Simulation of Finite Mixture Approach

We estimate the model without preference heterogeneity using the GMM two-step procedure described in Section 4.4. Specifically, we employ as regular instruments

Table 1: Demand Estimation with Unobserved Information Factor in the Three-type Monte Carlo Simulation

		Three-Type						Three-Type & RC		
		Regular IV			Chamberlain's IV			Chamberlain's IV		
param.	true	est.	st. err.	bias	est.	st. err.	bias	est.	st. err.	bias
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
β_0	1	1.02	(0.129)	0.02	1.00	(0.108)	0.00	1.00	(0.113)	-0.00
β_x	1	1.00	(0.055)	-0.00	1.00	(0.053)	-0.00	1.00	(0.058)	-0.00
α	1.5	1.50	(0.046)	-0.00	1.50	(0.045)	0.00	1.50	(0.050)	0.00
ϕ_A	0.15	0.16	(0.017)	0.01	0.15	(0.005)	-0.00	0.15	(0.007)	-0.00
ϕ_B	0.5	0.48	(0.033)	-0.02	0.50	(0.016)	-0.00	0.50	(0.021)	0.00
ϕ_C	0.35	0.34	(0.045)	-0.01	0.35	(0.018)	0.00	0.35	(0.023)	0.00
γ	-1.5	-3.13	(0.140)	-1.63	-1.50	(0.035)	0.00	-1.48	(0.051)	0.02
σ_γ	0.5							0.48	(0.049)	-0.02

Notes. All estimates are from 1,000 Monte Carlo simulations and the DGP includes three information types $\kappa = A, B, C$. Columns (3) to (8) are from a DGP where there is no individual heterogeneity in the distribution of preference parameters while columns (9) to (11) are from a DGP where individuals' valuation of the expected attribute follows a normal distribution with mean γ and standard deviation σ_γ. Regular instruments are the intercept, product characteristics x_{jt}, prices p_{jt}, cost shifter z_{jt}, and two information variables k_{1jt}, k_{2jt} with k_{1jt} observed and k_{2jt} estimated by a linear factor model. Chamberlain's IV replaces the latter two information variables in the instrument set.

constant, $x_{jt}, p_{jt}, z_{jt}, k_{1jt}$ and (estimated) k_{2jt} in the first stage. This preliminary instrument set is not by itself sufficient to separately identify all parameters in the model but the role of the first stage is to deliver initial estimates used to construct feasible optimal instruments.[14] We calculate the approximated Chamberlain's instruments for parameters γ and ϕ based on the first stage estimates following Reynaert and Verboven (2014).Then, we replace the information variables k_{1jt}, k_{2jt} in the set of regular instruments with Chamberlain's instruments in the second stage.

Table 1 displays the estimation results of a Monte Carlo simulation over the 1,000 datasets when three information types are present in the DGP. Columns (3)-(8) show that the bias of estimates using Chamberlain's instruments is smaller than those using regular instruments. Especially for the intercept, information parameters ϕ, and valuation of the expected attribute γ, the bias is significantly reduced when using Chamberlain's instruments. In Columns (9)-(11), we precisely estimate the information type shares and the random coefficients on the valuation of the expected attribute γ.

Appendix Table 5 displays the estimation results when the DGP includes two infor-

[14]When estimating the dataset with preference heterogeneity and containing only two information types B, C, we find that the first-stage estimates are usually heavily biased due to the lack of strong initial instruments, affecting the quality of Chamberlain's instruments in the second stage. To address this issue, we implement a three-step procedure by constructing Chamberlain's instrument twice.

mation types, B and C, and we include A, B, and C as candidate information types. The estimated mixture assigns zero weight to type A, correctly recovering that this type is not present in the DGP.[15]

6.3 Simulation of Moment Inequality Approach

For ease of computation, we modify the setup in Section 6.1 and simulate a single dataset consisting of $T = 2,500$ markets and $J = 10$ products. The indirect utility now excludes the price variable and is defined as

$$u^e_{ijt} = \beta_0 + \beta_x x_{jt} + \gamma g^e_{jt} + \xi_{jt} + \epsilon_{ijt}.$$

The data-generating process follows the standard three-type finite mixture setting, except that we swap the fractions of types A and B, setting $\phi_A = 0.5$ and $\phi_B = 0.15$. We estimate the moment inequalities in (15) using the information type A as the observed type $\tilde{\kappa}$ defined in Assumption (8). We select observations where type A's prediction errors e_{Ajt} are above the a-th quantile q_a (and conversely below the $(1-a)$-th quantile q_{1-a}) of its empirical distribution, with $a \in \{0.9, 0.8, \ldots, 0.5\}$. The selection conditions are defined as $Z^S_{\max jt} = \mathbb{1}\{e_{Ajt} \geq q_a\}$ and $Z^S_{\min jt} = \mathbb{1}\{e_{Ajt} \leq q_{1-a}\}$.

The moment inequalities in (12) rely on knowledge of the prediction error $e_{\tilde{\kappa}jt}$ for the extreme information type $\tilde{\kappa}$. In practice, the true extreme type may vary across observations and the selection instrument might misclassify observations where other types than $\tilde{\kappa}$. are extremal. Such misclassification introduces noise into the moment conditions, which weakens their ability to correctly detect violations and may lead to conservative confidence regions. Intuitively, the looser the selection threshold a, the greater the risk that the observed type is not sufficiently extreme relative to other types. On the other hand, including more observations increases the effective sample size, which improves statistical precision.

Our simulation setting allows us to observe the predictions of all information types when generating the data. This allows us to assess the misclassification errors and the bias-variance trade-offs of our threshold instrument.

Table 2 presents the simulation results. Our moment inequality approach yields informative and precise confidence bounds. We experiment with different selection

[15]By plotting the GMM objective function, we observe that the regular instruments lead to a problematic objective function with multiple local minima, as shown in Appendix Figure A3. However, the GMM objective function becomes well-behaved when using Chamberlain's instruments, as depicted in Appendix Figure A3, further supports the effectiveness of Chamberlain's instruments.

Table 2: Simulation Results with Different Selection Thresholds for $\tilde{\kappa} = A$

	β_x (1)	γ (2)	# Obs. (3)	# Misclass (4)	# Pts in CI (5)
True value	1	-1.5			
a					
0.9	[-0.14,1.94]	[-2.20,-0.15]	5 000	0	285
0.8	[0.14,1.77]	[-2.09,-0.48]	10 000	0	199
0.7	[0.14,1.66]	[-1.98,-0.48]	15 000	478	164
0.6	[0.25,1.61]	[-1.88,-0.58]	20 000	2 869	126
0.5	[0.36,1.49]	[-1.77,-0.69]	25 000	5 369	89

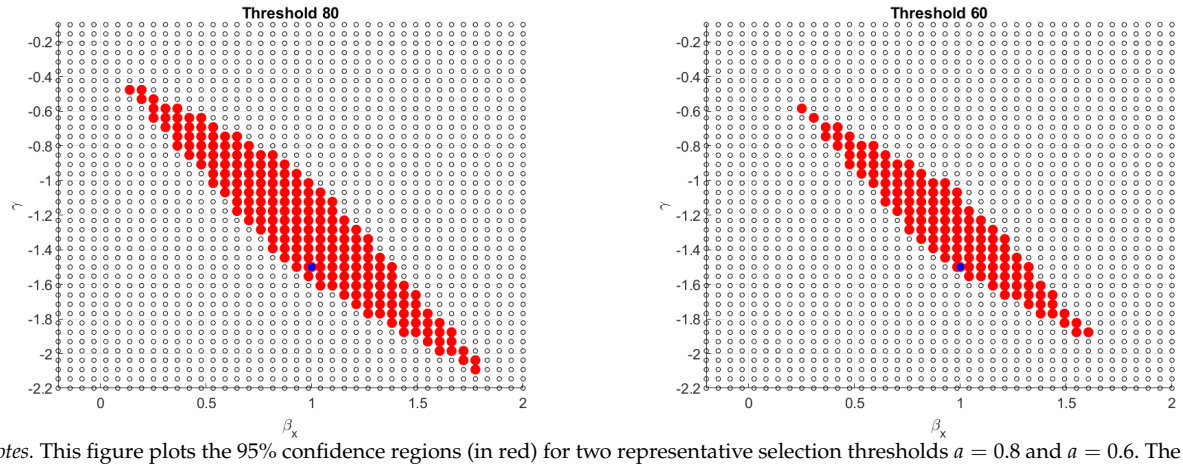
Notes. All estimates are from a large dataset of 25,000 observations. The data-generating process includes three information types $\kappa = A, B, C$ and we specify A as the observed extreme type $\tilde{\kappa}$ in the estimation. The instruments are the intercept, x_{jt}, and estimated expectations of the extreme type $\hat{\delta}^e_{Ajt}$. We present results for different selection thresholds a from 0.9 to 0.5. The selected samples' sizes are reported in column (3). Among the selected observations, we count the number of cases where their $\hat{\delta}^e_{Ajt}$ are not the most extreme of all type-specific predictions $\hat{\delta}^e_{\kappa jt}, \kappa = A, B, C$. This is reported in column (4). Both numbers of observations (3) and misclassification (4) are the sum of data points constructing the max and min inequalities (e.g., at 90% percentiles, we have 2,500 obs. for the max side and another 2,500 obs for the min side). In our simulation, the number of misclassified data points used in the estimation all comes from the max side of the inequality. For ease of computation, we do not estimate the intercept and fix it at its true value. We define a grid of 40 values for the other two parameters, with β_x ranging from -0.2 to 2 with a step of 0.055 and γ from -2.2 to -0.1 with a step of 0.0575. Extreme points of the 95% confidence set are reported in square brackets. The number of points $(\hat{\beta}_x, \hat{\gamma})$ inside the 95% confidence set is reported in column (5).

thresholds to illustrate the bias-variance trade-off inherent in our framework. Column (4) of Table 2 reports the number of misclassified observations included in the estimation sample. As expected, we find that as the sample size increases, the number of misclassified observations also increases, but their share, in general, remains small relative to the total estimation sample.

We find that, as the selection threshold decreases (i.e., more observations are included), the size of the confidence region continues to shrink, indicating that the sample size effect dominates the misclassification effect, and the inference is not overly conservative. Until a threshold of $a = 0.8$ there is no misclassification in the simulated data. When the threshold is reduced to $a = 0.7$, although there are around 3% of misclassified observations, the confidence region remains nearly identical to the one obtained with $a = 0.8$. When we decrease the threshold to $a = 0.5$, the full sample is used, and the misclassified observations account for over 20% of the estimation sample. Only in this case does the misclassification problem dominate and result in the exclusion of the true parameter from the confidence region.

Figure 1 illustrates the bias-variance trade-off by graphically showing the evolution of the confidence region. Supplemental evidence is also provided in Figure A4 that plots the estimated bounds for each parameter β_x and γ across different thresholds. This exercise

Figure 1: 95% Confidence Regions across Different Selection Thresholds ($\tilde{\kappa} = A$)



Notes. This figure plots the 95% confidence regions (in red) for two representative selection thresholds $a = 0.8$ and $a = 0.6$. The blue point indicates the true value. The confidence region shrinks when the sample size increases from $a = 0.8$ to $a = 0.6$. Although the sample with $a = 0.6$ contains around 14% misclassified observations, the estimated confidence region remains valid.

demonstrates that our moment inequality approach can deliver informative and valid inference, provided that (1) the sample size grows sufficiently fast relative to the number of misclassified observations, and (2) the share of misclassified observations remains bounded within the estimation sample.

7 Empirical Applications

We revisit two empirical papers where unobserved information structures play an important role. First, we revisit DM18, which uses individual-level data, and apply both our finite mixture and moment inequality approach. Second, we revisit the market-level data setting of GRV18 and apply our finite mixture approach.

7.1 What Do Exporters Know?

We revisit DM18, who study the information structure underlying Chilean firms' exporting decisions. Their approach relies on a minimal information set that is assumed to be known to all exporters and uses moment inequalities to obtain identified sets for the parameters governing fixed exporting costs. Our framework relaxes the common-information restriction by allowing firms to use heterogeneous prediction information sets. Rather than asking only "What do all exporters know?", we estimate the distribution of information types and quantify the fraction of firms that rely on different information sets when forming predictions about export profitability.

We use the same unbalanced panel studied in DM18, covering $N = 266$ Chilean firms in the food and chemical sectors over 1996–2005, and firm–destination export decisions to $J^c = 22$ (chemical) and $J^f = 34$ (food) destinations. Following DM18, the key uncertain object is the potential export revenue. The information variables highlighted in DM18 include distance to the destination, lagged domestic revenue, lagged aggregate export revenue to the destination, the lagged revenue shifter, and the lagged number of exporters to the destination. We take the minimal set $\{dist_j, r_{iht-1}, R_{jt-1}\}$ as a baseline type and allow additional types to incorporate the lagged revenue shifter α_{jt-1} and/or the lagged number of exporters N_{jt-1}, yielding four candidate information types.

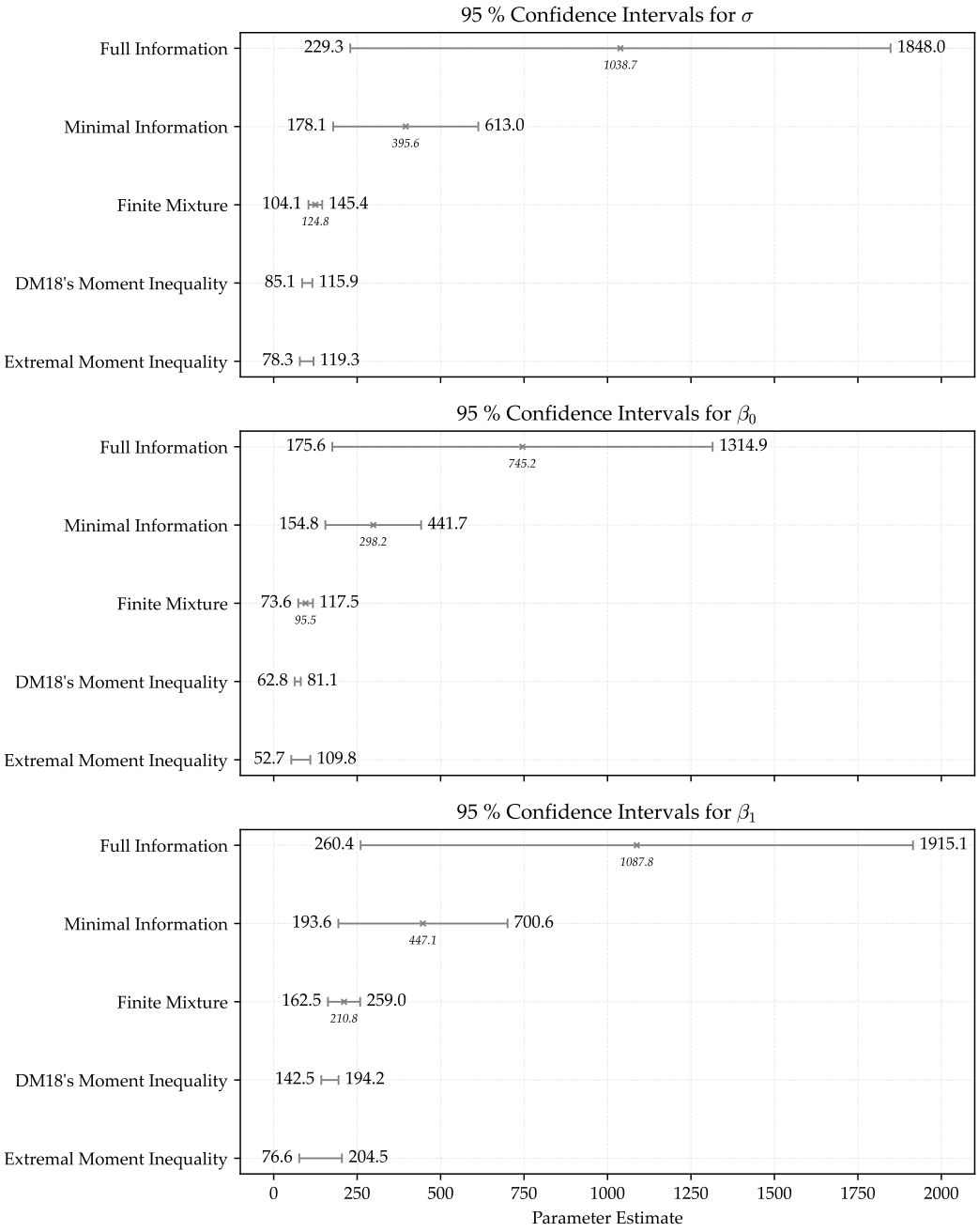
We estimate a finite-mixture version of the export decision model by maximum likelihood, where the mixture weights ϕ capture the population shares of firms using each information type. In parallel, we report moment-inequality estimates based on the minimal-information benchmark and on our moment inequality approach. Estimation details—including the construction of destination-year revenue shifters, the computation of type-specific predicted revenues, and the implementation of the moment inequalities—are provided in Appendix B.

Figure 2 compares parameter estimates across the perfect-foresight plug-in benchmark, the minimal-information plug-in benchmark, the DM18 inequality estimator, and our information-robust estimators. We confirm that naive plug-in approaches can lead to substantial bias. Our extremal moment-inequality approach delivers an identified set that overlaps with the DM18 set.[16] The finite-mixture confidence intervals overlap with both inequality approaches.

Table 3 reports the estimated information-type shares. We find substantial heterogeneity in firms' prediction information sets: 15% of firms are consistent with the minimal-information type, 56% incorporate information on the lagged number of exporters, and the type that adds only the lagged revenue shifter receives negligible weight. We also estimate that 29% of firms use the full set of observed information variables. These results illustrate how the finite-mixture approach recovers not only preference parameters but also the distribution of information types that rationalizes observed exporting behavior.

[16] Our identified set is wider because we aggregate individual choices into market shares, which reduces the informativeness of the data.

Figure 2: 95% Confidence Intervals for the Chemical Sector



Notes. All estimates are reported in thousands of year 2000 US$, and their magnitudes scale proportionally with the parameter η, which is set to 5, following DM18. For each parameter σ, β_0, and β_1, we first plot 95% confidence intervals corresponding to three maximum likelihood estimators (MLE), with point estimates indicated by an 'x' at the center of each interval. The "Full Information" corresponds to the "Perfect Foresight" MLE in DM18, which assumes that firms' expectations coincide with realized outcomes, i.e., $\mathbb{E}[r_{ijt}|\mathcal{I}ijt] = rijt$. The "Minimal Information" replicates the specification in DM18 where firms form expectations using the minimal information set. The third row presents confidence intervals from our finite mixture model with four information types. Then, we plot the extreme points of the 95% confidence sets for the two moment inequality estimators. The "Finite Mixture" and the two moment inequality estimators are robust to unobserved information structure.

Table 3: Finite Mixture Estimates for Entry Decision in the Chemical Sector

Parameters	σ	β_0	β_1	ϕ_A	ϕ_B	ϕ_C	ϕ_D
	124.8	95.5	210.8	0.15	1.17E-12	0.56	0.29
	(11)	(11)	(25)	(0.048)	(0.092)	(0.078)	(–)

Notes. All the preference estimates $(\sigma, \beta_0, \beta_1)$ are reported in thousands of year 2000 US$, and their values scale proportionally with η, which is set equal to 5, as in DM18. The finite mixture model consists of four information types $\kappa = A, B, C, D$ and the estimated share of the information type D is calculated as $\phi_D = 1 - \phi_A - \phi_B - \phi_C$. Bootstrap standard errors are computed as in DM18 and reported in parentheses.

7.2 Consumer Valuation of Fuel Costs

A key challenge in the literature on how purchases of energy-consuming durables respond to future operating costs (e.g., Hausman, 1979) is that researchers rarely observe consumers' expectations about future energy expenses at the time of purchase. A common approach is therefore to specify an expectation-formation rule for future expenses and plug the implied expectation into the purchase model. When consumers' actual predictions differ from the researcher's maintained rule, such plug-in approaches are subject to the misspecified-information bias discussed in Section 3.

In this application, we use our framework to allow for heterogeneous prediction information sets when forming expected future expenses and to quantify how alternative prediction types affect inferred valuation of energy. We revisit the estimation framework of GRV18, who study the valuation of automobile fuel costs in the EU market using a demand model with rich preference heterogeneity that combines random coefficients with an empirical mileage distribution. Their estimates suggest that consumers undervalue expected fuel costs at the time of purchase, but the framework does not account for information heterogeneity. We recast the problem in our framework and estimate a finite mixture over observable information types. This delivers fuel-cost valuation estimates consistent with a richer information structure and yields direct estimates of the prevalence of different information types in the data.

Model. A consumer i chooses among vehicle products indexed by model j and engine variant k in market t (defined as a year–country pair). For simplicity, we suppress the market subscript t. Expected utility is

$$u^e_{ijk} = x'_{jk}\beta^x_i - \alpha_i\left(p_{jk} + \gamma g^e_{ijk}\right) + \xi_{jk} + \epsilon_{ijk},$$

where x_{jk} is a vector of observed characteristics, p_{jk} is price, g^e_{ijk} denotes consumer i's expected fuel cost, ξ_{jk} is an unobserved product attribute, and ϵ_{ijk} a type-I extreme value

error term. The coefficients β_i^x and α_i capture heterogeneity in tastes for characteristics and in the marginal utility of income.

Expected fuel costs incorporate heterogeneity in mileage. Following GRV18, we write the present discounted value of expected fuel expenditures as

$$g_{ijk}^e = \rho \, \beta_i^m \, e_{jk} \, f_k,$$

where β_i^m is consumer i's annual mileage, e_{jk} is fuel consumption, f_k is the fuel price, and

$$\rho \equiv \sum_{s=1}^{S} (1+r)^{-s}$$

is the capitalization factor determined by vehicle lifetime S and interest rate r. The parameter γ measures the degree to which consumers internalize future fuel costs at purchase: $\gamma < 1$ corresponds to under-weighting g_{ijk}^e relative to the contemporaneous price p_{jk}.

GRV18 assumes that consumers have full information about future fuel costs. We relax this assumption and investigate what information consumers use to form their expectations g_{ijk}^e. We assume consumers know their mileage β_i^m and consider the observed fuel efficiency e_{jk} and fuel prices f_k as the information variables to construct a mixture model of three information types, indexed by $\kappa = A, B, C$, where the information set of each type is $\mathcal{W}_A = \{e_{jk}\}$, $\mathcal{W}_B = \{f_k\}$, and $\mathcal{W}_C = \{e_{jk}, f_k\}$, respectively. We denote the product $e_{jk} f_k$ as the uncertain attribute g_{jk} in our model. The expectation of each information type is $g_{\kappa jk}^e = \mathbb{E}[e_{jk} f_k | \mathcal{W}_\kappa]$. The information type C corresponds to the perfect foresight type that uses both fuel efficiency and fuel price to predict fuel costs.

The market share for product (j,k) is

$$s_{jk} = \sum_{\kappa=1}^{|\mathcal{K}|} \phi_\kappa \int \frac{\exp\left(x_{jk}'\beta_i^x - \alpha_i p_{jk} - \alpha_i \gamma\rho \, \beta_i^m \, g_{\kappa jk}^e + \xi_{jk}\right)}{1 + \sum_{(j',k')} \exp\left(x_{j'k'}'\beta_i^x - \alpha_i p_{j'k'} - \alpha_i \gamma\rho \, \beta_i^m \, g_{\kappa j'k'}^e + \xi_{j'k'}\right)} \, dF_\beta(\beta_i;\theta), \qquad (16)$$

where $\beta_i \equiv (\beta_i^{x\prime}, \alpha_i, \gamma\rho\beta_i^m)'$ collects the heterogeneous coefficients entering utility. The component β_i^x follows a parametric distribution indexed by θ (means and variance–covariance parameters). The remaining heterogeneity is introduced through empirical draws: α_i depends on the empirical income distribution, and β_i^m is drawn from the empirical mileage distribution and scaled by $\gamma\rho$. The parameter vector is $\Theta \equiv (\theta', \phi')'$.

Estimation. We estimate the market-share system in (16) using a panel of T markets. The vector of observed characteristics x_{jkt} includes horsepower, size and height, an indicator

for foreign production, and a diesel indicator interacted with country dummies. Following GRV18, we restrict random coefficients on characteristics to $\beta_i^x = \bar{\beta}^x + \Sigma^x \nu_i^x$, where $\bar{\beta}^x$ collects mean valuations, Σ^x is diagonal with standard deviations σ^x on the diagonal, and $\nu_i^x \sim \mathcal{N}(0, I)$. The marginal utility of income scales with market income y_t as $\alpha_i = \alpha / y_t$. Individual mileage β_i^m is drawn from the empirical mileage distribution used in GRV18. The parameter vector is

$$\Theta = \left(\bar{\beta}^{x\prime}, \sigma^{x\prime}, \alpha, \gamma\rho, \phi_A, \phi_B, \phi_C\right)'.$$

Finally, unobserved quality is specified as

$$\xi_{jkt} = \xi_j + \xi_t + \tilde{\xi}_{jkt},$$

where ξ_j are model fixed effects and ξ_t are market fixed effects (implemented as country-specific linear and quadratic time trends). The model is estimated by GMM using product-level instruments z_{jkt} and moment conditions

$$\mathbb{E}\left[z_{jkt}\, \tilde{\xi}_{jkt}\right] = 0,$$

where $\tilde{\xi}_{jkt}$ is the structural error after partialling out the included fixed effects. Equivalently, the excluded instruments are orthogonal to the unobserved quality component conditional on the included fixed effects.

We approximate the conditional expectation operator $g^e_{\kappa jk} = \mathbb{E}[e_{jk} f_k \mid \mathcal{W}_\kappa]$ using a random-forest predictor. This flexibly captures nonlinear relationships between the information variables and expected fuel costs, as illustrated in Appendix Figure A5. In our framework, this first-stage nonparametric prediction provides an empirical estimate of the conditional-expectation mapping and therefore serves as the empirical counterpart to Assumption 3.

We first estimate the mixture model in Equation (16) without random coefficients using the standard two-step GMM. In the first stage, our instruments z_t include observed product attributes x_{jkt}, cost shifters, BLP instruments and the realized fuel costs g_{jkt}. Then we follow Reynaert and Verboven (2014) to calculate the approximated Chamberlain's instruments for the fraction parameters ϕ_κ and the undervaluation coefficient $\gamma\rho$ in the second stage.

When allowing for preference heterogeneity, the standard two-step GMM procedure yields imprecise estimates in the mixture model (16). We therefore adopt a one-step estimator that continuously updates the Chamberlain instruments during optimization, following Bourreau et al. (2021). We minimize the GMM objective using Newton's method

with numerically computed gradients.

Results. Table 4 reports parameter estimates for the GRV18 specifications in columns (1)–(3) and for the corresponding finite-mixture versions in columns (4)–(6). The key implication is that the data is difficult to reconcile with a perfect-foresight benchmark: the estimated share of the full-information type is essentially zero in the logit and RC Logit I specifications, and remains small in RC Logit II ($\hat{\phi}_C = 0.10(0.04)$). The small standard errors on the estimated share of perfect-foresight consumers suggest that the data is consistent with a setting where these consumers do not play a signicant role in the car market. Moreover, the fact that the full-information type receives little weight is consistent with external evidence that consumers' ex-ante and ex-post fuel costs differ substantially (see for instance, Levinson and Sager, 2023).

The estimated mixture weights reveal substantial heterogeneity in consumers' expectations about fuel costs. Across specifications, roughly one-half to two-thirds of consumers rely primarily on fuel efficiency (type A), with the remainder relying on fuel prices (type B). Fuel costs therefore enter choice through heterogeneous prediction rules rather than a common full-information expectation.

Allowing for heterogeneous predictions substantially alters the inferred valuation of future fuel costs. Relative to GRV18, the mixture specification yields markedly lower present-value weights: the implied future-valuation parameter $\hat{\gamma}$ declines from approximately 0.77–0.91 under the full-information specification to 0.51, 0.43, and 0.22 in columns (4)–(6). These results demonstrate that imposing a plug-in full-information structure attributes cross-sectional variation in beliefs to preferences. As a consequence, standard specifications overstate attention to future operating costs and may generate systematically distorted policy counterfactuals.

8 Conclusion

This paper introduces two novel econometric approaches to estimating discrete choice models in the presence of unobserved heterogeneous information sets. We illustrate the significance of accounting for unobserved information heterogeneity through empirical applications and simulations. Our analysis of firm export decisions confirms that our methods successfully find estimates similar to previous approaches but under very different assumptions. In the context of consumer vehicle purchases, our results reject the commonly assumed perfect foresight about fuel costs and reveal substantial heterogeneity in how consumers form expectations.

Table 4: Parameter Estimates for Alternative Demand Models

	Table 3 in Grigolon et al. (2018)			Finite Mixture		
	Logit	RC Logit I	RC Logit II	Logit	RC Logit I	RC Logit II
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. Mean Valuations						
Price/inc. ($-\alpha$)	-4.52	-6.22	-5.33	-2.43	-4.40	-4.97
	(0.19)	(0.22)	(0.21)	(0.21)	(0.21)	(0.28)
Fuel costs/inc. ($-\alpha\gamma\rho$)	-39.03	-46.48	-47.11	-12.71	-18.41	-11.07
	(1.41)	(0.94)	(9.22)	(0.33)	(0.64)	(0.50)
Power (kW/100)	2.28	2.6	0.25	1.13	2.23	-1.87
	(0.14)	(0.17)	(0.61)	(0.14)	(0.14)	(0.18)
Size (cm^2/10k)	13.25	16.69	16.77	14.36	15.76	14.39
	(0.44)	(0.48)	(2.02)	(0.41)	(0.42)	(1.40)
Height (cm/100)	3.00	4.45	5.19	3.04	3.85	5.59
	(0.30)	(0.32)	(0.33)	(0.26)	(0.28)	(0.18)
Foreign	-0.83	-0.75	-0.89	-0.94	-0.84	-1.09
	(0.02)	(0.02)	(0.04)	(0.02)	(0.02)	(0.05)
Panel B. Fractions of Information Types						
Fuel efficiency (ϕ_A)				0.68	0.66	0.51
				(0.02)	(0.02)	(0.04)
Fuel price (ϕ_B)				0.32	0.34	0.39
				(0.03)	(0.03)	(0.04)
Perfect foresight (ϕ_C)				0.00	0.00	0.10
				(0.04)	(0.03)	(0.04)
Panel C. Standard Deviations of Valuations						
Power (kW/100)			1.95			2.70
			(0.25)			(0.18)
Size (cm^2/10k)			4.31			6.04
			(2.04)			(1.24)
Foreign			0.49			1.56
			(0.43)			(0.06)
Mileage distribution	No	Yes	Yes	No	Yes	Yes
Panel D. Valuations of Future Fuel Costs						
Fuel costs/price ($\gamma\rho$)	8.63	7.47	8.84	5.24	4.19	2.23
	(0.55)	(0.24)	(1.77)	(0.35)	(0.14)	(0.11)
Future val. γ ($r = 6\%$)	0.89	0.77	0.91	0.51	0.41	0.22
	(0.06)	(0.02)	(0.18)	(0.03)	(0.01)	(0.01)

Notes. This table reproduces estimates from GRV18 in columns (1)-(3) and applies the finite mixture model with three information types in columns (4)-(6). Standard errors are reported in parentheses.

Our methodological contributions extend existing work on discrete choice under uncertainty by introducing new tools that can be readily implemented in empirical research. The finite mixture model offers a tractable solution for recovering informational heterogeneity using standard choice data. At the same time, the moment inequality approach provides a more robust alternative in settings where the researcher cannot specify all information variables. Given the increasing relevance of policy interventions that aim to influence choice through information provision, our framework is valuable for assessing how informational heterogeneity affects counterfactual predictions.

Future research may explore extensions of our methods to dynamic decision-making settings and contexts with dynamic learning. Additionally, further work is needed to understand how these approaches can be applied in high-dimensional settings where the set of potential information variables is large. By bridging the gap between observed choices and unobserved expectations, our work advances the empirical toolkit available to researchers studying economic decision-making under uncertainty.

References

Abaluck, Jason and Abi Adams-Prassl (2021) "What do Consumers Consider Before They Choose? Identification from Asymmetric Demand Responses*," *The Quarterly Journal of Economics*, 136 (3), 1611–1663, 10.1093/qje/qjab008.

Abaluck, Jason and Giovanni Compiani (2020) "A Method to Estimate Discrete Choice Models that is Robust to Consumer Search," Working Paper 26849, National Bureau of Economic Research, 10.3386/w26849.

Agarwal, Nikhil and Paulo J Somaini (2022) "Demand analysis under latent choice constraints,"Technical report, National Bureau of Economic Research.

Ahn, Hyungtaik and Charles F Manski (1993) "Distribution theory for the analysis of binary choice under uncertainty with nonparametric estimation of expectations," *Journal of Econometrics*, 56 (3), 291–321, https://doi.org/10.1016/0304-4076(93)90123-M.

Allcott, Hunt and Christopher Knittel (2019) "Are Consumers Poorly Informed about Fuel Economy? Evidence from Two Experiments," *American Economic Journal: Economic Policy*, 11 (1), 1–37, 10.1257/pol.20170019.

Andrews, Donald W. K. and Xiaoxia Shi (2013) "Inference Based on Conditional Moment Inequalities," *Econometrica*, 81 (2), 609–666, https://doi.org/10.3982/ECTA9370.

Andrews, Isaiah, Jonathan Roth, and Ariel Pakes (2023) "Inference for Linear Conditional Moment Inequalities," *The Review of Economic Studies*, 90 (6), 2763–2791, 10.1093/restud/rdad004.

Arcidiacono, Peter, V. Joseph Hotz, Arnaud Maurel, and Teresa Romano (2020) "Ex Ante Returns and Occupational Choice," *Journal of Political Economy*, 128 (12), 4475–4522, 10.1086/710559.

Aucejo, Esteban and Jonathan James (2021) "The Path to College Education: The Role of Math and Verbal Skills," *Journal of Political Economy*, 129 (10), 2905–2946, 10.1086/715417.

Barahona, Nano, Cristóbal Otero, and Sebastián Otero (2023) "Equilibrium Effects of Food Labeling Policies," *Econometrica*, 91 (3), 839–868, https://doi.org/10.3982/ECTA19603.

Barseghyan, Levon, Maura Coughlin, Francesca Molinari, and Joshua C. Teitelbaum (2021) "Heterogeneous Choice Sets and Preferences," *Econometrica*, 89 (5), 2015–2048, https://doi.org/10.3982/ECTA17448.

Bergemann, Dirk, Benjamin Brooks, and Stephen Morris (2022) "Counterfactuals with Latent Information," *American Economic Review*, 112 (1), 343–68, 10.1257/aer.20210496.

Berry, Steven and Panle Jia (2010) "Tracing the Woes: An Empirical Analysis of the Airline

Industry," *American Economic Journal: Microeconomics*, 2 (3), 1–43, 10.1257/mic.2.3.1.

Berry, Steven, James Levinsohn, and Ariel Pakes (1995) "Automobile Prices in Market Equilibrium," *Econometrica*, 63 (4), 841–890, http://www.jstor.org/stable/2171802.

Berry, Steven T. (1994) "Estimating Discrete-Choice Models of Product Differentiation," *The RAND Journal of Economics*, 25 (2), 242–262, http://www.jstor.org/stable/2555829.

Berry, Steven T. and Philip A. Haile (2014) "Identification in Differentiated Products Markets Using Market Level Data," *Econometrica*, 82 (5), 1749–1797, https://doi-org.gorgone.univ-toulouse.fr/10.3982/ECTA9027.

Bonhomme, Stéphane, Thibaut Lamadon, and Elena Manresa (2022) "Discretizing Unobserved Heterogeneity," *Econometrica*, 90 (2), 625–643, https://doi.org/10.3982/ECTA15238.

Bourreau, Marc, Yutec Sun, and Frank Verboven (2021) "Market Entry, Fighting Brands, and Tacit Collusion: Evidence from the French Mobile Telecommunications Market," *American Economic Review*, 111 (11), 3459–99, 10.1257/aer.20190540.

Brown, Zach Y. (2019) "An Empirical Model of Price Transparency and Markups in Health Care," working paper, National Bureau of Economic Research, https://websites.umich.edu/~zachb/zbrown_empirical_model_price_transparency.pdf.

Brown, Zach Y, Christopher Hansman, Jordan Keener, and Andre F Veiga (2023) "Information and Disparities in Health Care Quality: Evidence from GP Choice in England," Working Paper 31033, National Bureau of Economic Research, 10.3386/w31033.

Brown, Zach Y. and Jihye Jeon (2024) "Endogenous Information and Simplifying Insurance Choice," *Econometrica*, 92 (3), 881–911, https://doi.org/10.3982/ECTA18555.

Chamberlain, Gary (1987) "Asymptotic efficiency in estimation with conditional moment restrictions," *Journal of Econometrics*, 34 (3), 305–334, https://doi.org/10.1016/0304-4076(87)90015-7.

Cox, Gregory and Xiaoxia Shi (2023) "Simple Adaptive Size-Exact Testing for Full-Vector and Subvector Inference in Moment Inequality Models," *The Review of Economic Studies*, 90 (1), 201–228, 10.1093/restud/rdac015.

Cunha, Flavio and James J. Heckman (2007) "Identifying and Estimating the Distributions of Ex Post and Ex Ante Returns to Schooling," *Labour Economics*, 14 (6), 870–893, https://doi.org/10.1016/j.labeco.2007.06.002, Education and Risk.

Cunha, Flavio, James Heckman, and Salvador Navarro (2005) "Separating uncertainty from heterogeneity in life cycle earnings," *Oxford Economic Papers*, 57 (2), 191–261, 10.1093/oep/gpi019.

Dickstein, Michael J, Jihye Jeon, and Eduardo Morales (2024) "Patient Costs and Physicians' Information," Working Paper 32014, National Bureau of Economic Research, 10.3386/w32014.

Dickstein, Michael J and Eduardo Morales (2018) "What do Exporters Know?," *The Quarterly Journal of Economics*, 133 (4), 1753–1801, 10.1093/qje/qjy015.

D'Haultfœuille, Xavier, Arnaud Maurel, and Yichong Zhang (2018) "Extremal quantile regressions for selection models and the black–white wage gap," *Journal of Econometrics*, 203 (1), 129–142, https://doi.org/10.1016/j.jeconom.2017.11.004.

Einav, Liran, Amy Finkelstein, and Mark R. Cullen (2010) "Estimating Welfare in Insurance Markets Using Variation in Prices," *The Quarterly Journal of Economics*, 125 (3), 877–921, 10.1162/qjec.2010.125.3.877.

Gandhi, Amit, Zhentong Lu, and Xiaoxia Shi (2023) "Estimating demand for differentiated products with zeroes in market share data," *Quantitative Economics*, 14 (2), 381–418, https://doi.org/10.3982/QE1593.

Grigolon, Laura, Mathias Reynaert, and Frank Verboven (2018) "Consumer Valuation of Fuel Costs and Tax Policy: Evidence from the European Car Market," *American Economic Journal: Economic Policy*, 10 (3), 193–225, 10.1257/pol.20160078.

Hausman, Jerry A. (1979) "Individual Discount Rates and the Purchase and Utilization of Energy-Using Durables," *The Bell Journal of Economics*, 10 (1), 33–54, http://www.jstor.org/stable/3003318.

Houmark, Mikkel Aagaard, Victor Ronda, and Michael Rosholm (2024) "The Nurture of Nature and the Nature of Nurture: How Genes and Investments Interact in the Formation of Skills," *American Economic Review*, 114 (2), 385–425, 10.1257/aer.20220456.

Levinson, Arik and Lutz Sager (2023) "Who Values Future Energy Savings? Evidence from American Drivers," *Journal of the Association of Environmental and Resource Economists*, 10 (3), 717–751.

Manski, Charles F. (1993) *2. Adolescent Econometricians: How Do Youth Infer the Returns to Schooling?*, 43–60, Chicago: University of Chicago Press, doi:10.7208/9780226110646-005.

——— (2004) "Measuring Expectations," *Econometrica*, 72 (5), 1329–1376, http://www.jstor.org/stable/3598892.

Molinari, Francesca (2020) "Microeconometrics with partial identification," in *Handbook of Econometrics, Volume 7A*, 7 of Handbook of Econometrics, 355–486: Elsevier, https://doi.org/10.1016/bs.hoe.2020.05.002.

Nevo, Aviv, John L Turner, and Jonathan W Williams (2016) "Usage-based pricing and demand for residential broadband," *Econometrica*, 84 (2), 411–443.

Porcher, Charly, Eduardo Morales, and Thomas Fujiwara (2024) "Measuring Information Frictions in Migration Decisions: A Revealed-Preference Approach," Working Paper 32413, National Bureau of Economic Research, 10.3386/w32413.

Reynaert, Mathias and Frank Verboven (2014) "Improving the performance of random coefficients demand models: The role of optimal instruments," *Journal of Econometrics*, 179 (1), 83–98, https://doi.org/10.1016/j.jeconom.2013.12.001.

Vatter, Benjamin (2024) "Quality disclosure and regulation: Scoring design in medicare advantage," working paper, `https://ssrn.com/abstract=4250361`.

Wang, Ao (2023) "Sieve BLP: A semi-nonparametric model of demand for differentiated products," *Journal of Econometrics*, 235 (2), 325–351, https://doi.org/10.1016/j.jeconom.2022.04.002.

Appendices

Appendix A Proofs

A.1 Proof of Partial Identification

Proposition 2 (Partial identification of γ)**.** *For simplicity, set $\beta = 0$, $\xi_{jt} = 0$, and suppose utility depends on the latent expectation g^e_{ijt} only through the index γg^e_{ijt}. Assume the (multinomial) logit choice probabilities satisfy, for $j \in \{0, 1, \ldots, J\}$,*

$$\Pr(d_{ijt} = 1 \mid \mathcal{W}_{it}) = \frac{\exp(\gamma g^e_{ijt})}{\sum_{k=0}^{J} \exp(\gamma g^e_{ikt})}. \tag{17}$$

Suppose agents only use one information variable $\mathcal{W}_{it} = \{\mathcal{K}_{ijt}\}$ the researcher observes $(g_{ijt}, \mathcal{K}_{ijt})$ and these satisfy the measurement system

$$g_{ijt} = g^e_{ijt} + e_{ijt}, \qquad \mathcal{K}_{ijt} = g^e_{ijt} + v_{ijt}, \tag{18}$$

with $\mathbb{E}[e_{ijt} \mid g^e_{ijt}] = 0$ and finite second moments. Let

$$\sigma^2_g \equiv \mathrm{Var}(g_{ijt}), \quad \sigma^2_{\mathcal{K}} \equiv \mathrm{Var}(\mathcal{K}_{ijt}), \quad \sigma_{g\mathcal{K}} \equiv \mathrm{Cov}(g_{ijt}, \mathcal{K}_{ijt}),$$

be identified from the joint distribution of the observables $(g_{ijt}, \mathcal{K}_{ijt})$. If $\mathrm{Cov}(g^e_{ijt}, v_{ijt})$ is unrestricted, then γ is not point identified.

Proof. Step 1 (Observable second moments). From (18) and $\mathbb{E}[e_{ijt} \mid g^e_{ijt}] = 0$, we have $\mathrm{Cov}(g^e_{ijt}, e_{ijt}) = 0$. Hence,

$$\mathrm{Var}(g_{ijt}) = \mathrm{Var}(g^e_{ijt}) + \mathrm{Var}(e_{ijt}) \;=\; \sigma^2_{g^e} + \sigma^2_e, \tag{19}$$

$$\mathrm{Var}(\mathcal{K}_{ijt}) = \mathrm{Var}(g^e_{ijt}) + \mathrm{Var}(v_{ijt}) + 2\mathrm{Cov}(g^e_{ijt}, v_{ijt}) \;=\; \sigma^2_{g^e} + \sigma^2_v + 2\sigma_{g^e,v}, \tag{20}$$

$$\mathrm{Cov}(g_{ijt}, \mathcal{K}_{ijt}) = \mathrm{Cov}(g^e_{ijt} + e_{ijt},\, g^e_{ijt} + v_{ijt}) \;=\; \mathrm{Var}(g^e_{ijt}) + \mathrm{Cov}(g^e_{ijt}, v_{ijt}) \;=\; \sigma^2_{g^e} + \sigma_{g^e,v}, \tag{21}$$

where $\sigma^2_{g^e} \equiv \mathrm{Var}(g^e_{ijt})$, $\sigma^2_e \equiv \mathrm{Var}(e_{ijt})$, $\sigma^2_v \equiv \mathrm{Var}(v_{ijt})$, and $\sigma_{g^e,v} \equiv \mathrm{Cov}(g^e_{ijt}, v_{ijt})$. The left-hand sides of (19)–(21) are identified from observables.

Step 2 (Non-identification of the scale of g^e). The system (19)–(21) provides three

restrictions on the four unknowns $(\sigma^2_{g^e}, \sigma^2_e, \sigma^2_v, \sigma_{g^e,v})$. Because $\sigma_{g^e,v}$ is unrestricted, $\sigma^2_{g^e}$ is not uniquely determined by $(\sigma^2_g, \sigma^2_{\mathcal{K}}, \sigma_{g\mathcal{K}})$.

Step 3 (Logit depends on (γ, g^e) only through γg^e). By (17), for any $c > 0$,

$$\frac{\exp(\gamma g^e_{ijt})}{\sum_{k=0}^{J} \exp(\gamma g^e_{ikt})} = \frac{\exp((\gamma/c)\,(c g^e_{ijt}))}{\sum_{k=0}^{J} \exp((\gamma/c)\,(c g^e_{ikt}))}. \tag{22}$$

Thus rescaling (γ, g^e) as $(\gamma', g^{e\prime}) = (\gamma/c,\ c g^e)$ leaves choice probabilities (and hence the conditional distribution of d_{it} given g^e) unchanged.

Step 4 (Two observationally equivalent structures). Fix any $c \neq 1$. Choose an alternative variance $\sigma^{2\prime}_{g^e} = c^2 \sigma^2_{g^e}$ and set $\gamma' = \gamma/c$. Define

$$\sigma^{2\prime}_e \equiv \sigma^2_g - \sigma^{2\prime}_{g^e}, \qquad \sigma'_{g^e,v} \equiv \sigma_{g\mathcal{K}} - \sigma^{2\prime}_{g^e}, \qquad \sigma^{2\prime}_v \equiv \sigma^2_{\mathcal{K}} - \sigma^{2\prime}_{g^e} - 2\sigma'_{g^e,v}.$$

For any c such that these variances are nonnegative, the primed parameters satisfy (19)–(21) with the same observable moments $(\sigma^2_g, \sigma^2_{\mathcal{K}}, \sigma_{g\mathcal{K}})$. Therefore, the joint distribution of observables $(g_{ijt}, \mathcal{K}_{ijt})$ is unchanged, while (22) ensures that the implied logit choice probabilities are unchanged as well. Since $\gamma' \neq \gamma$ for $c \neq 1$, there exist at least two distinct values of γ consistent with the same observable distribution. Hence, γ is not point-identified, i.e., it may be partially identified. ■

A.2 Proof of Proposition 1

Proof. Let $\tilde{g}^e_{jt} \equiv (g^e_{1,jt}, \ldots, g^e_{|\mathcal{K}|,jt})' \in \mathbb{R}^{|\mathcal{K}|}$ collect the type-specific predictions for option j in market t. Define the type indicator vector $b_{\kappa(i)} \in \mathbb{R}^{|\mathcal{K}|}$ as the $\kappa(i)$-th canonical basis vector.[17] Then $g^e_{\kappa(i),jt} = b'_{\kappa(i)} \tilde{g}^e_{jt}$. With individual-specific scale γ_i, define the random-coefficient vector on the prediction component as

$$\tilde{\gamma}_i \equiv \gamma_i\, b_{\kappa(i)} \in \mathbb{R}^{|\mathcal{K}|}.$$

It follows that

$$\gamma_i g^e_{\kappa(i),jt} \;=\; \gamma_i\, b'_{\kappa(i)} \tilde{g}^e_{jt} \;=\; \tilde{g}^{e\prime}_{jt} \tilde{\gamma}_i.$$

Stack the regular characteristics with heterogeneous tastes $\tilde{x}^{(2)}_{jt}$ and the prediction vector

[17] For example, if $\kappa(i) = 1$ then $b_{\kappa(i)} = (1, 0, \ldots, 0)'$.

$\tilde{g}^e_{jt}$ into

$$x^{(2)}_{jt} \equiv (\tilde{x}^{(2)}_{jt},\, \tilde{g}^e_{jt}), \qquad \beta_i \equiv (\beta^{(2)}_i,\, \tilde{\gamma}_i).$$

Substituting into (10) yields

$$u^e_{ijt} = x^{(1)\prime}_{jt}\beta - \alpha p_{jt} + x^{(2)\prime}_{jt}\beta_i + \xi_{jt} + \epsilon_{ijt},$$

which is a standard random utility model with random coefficients β_i. The information structure implies that the prediction-component coefficient $\tilde{\gamma}_i$ lies on one of the finitely many directions $\{b_\kappa\}_{\kappa\in\mathcal{K}}$, scaled by γ_i, completing the isomorphism. ■

A.3 Proof of Theorem 2

In this section, we show how the identification results of Berry and Haile (2014) and Wang (2023) apply to our model to provide the proof for Theorem 2 in three steps.

Let $(\beta^{(2)}_i, \gamma_i) \in \mathbb{R}^{|\tilde{x}^{(2)}|} \times \mathbb{R}$ have joint distribution $\tilde{F}_{\beta^{(2)},\gamma}$. Let the information type $\kappa(i) \in \mathcal{K}$ be independent of $(\beta^{(2)}_i, \gamma_i)$ with $\Pr(\kappa(i) = \kappa) = \phi_\kappa$. Define the random coefficient vector $\beta_i = (\beta^{(2)}_i, \tilde{\gamma}_i)$, where $\tilde{\gamma}_i = \gamma_i b_{\kappa(i)}$. The induced distribution of $(\beta^{(2)}_i, \tilde{\gamma}_i)$, denoted $F_{\beta^{(2)},\tilde{\gamma}}$, is therefore a finite mixture. Formally, for any Borel set $\mathcal{B} \subseteq \mathbb{R}^{|\tilde{x}^{(2)}|} \times \mathbb{R}^{|\mathcal{K}|}$,

$$F_{\beta^{(2)},\tilde{\gamma}}(\mathcal{B}) = \Pr\big((\beta^{(2)}_i, \tilde{\gamma}_i) \in \mathcal{B}\big) = \sum_{\kappa=1}^{|\mathcal{K}|} \phi_\kappa \int \mathbf{1}\Big\{(\beta^{(2)}_i, \gamma_i b_\kappa) \in \mathcal{B}\Big\}\, d\tilde{F}_{\beta^{(2)},\gamma}(\beta^{(2)}_i, \gamma_i).$$

The support of the joint random-coefficient distribution $F_{\beta^{(2)},\tilde{\gamma}}$ lies on the union of $|\mathcal{K}|$ coordinate subspaces,

$$\mathcal{S} = \bigcup_{\kappa=1}^{|\mathcal{K}|} \Big(\mathbb{R}^{|\tilde{x}^{(2)}|} \times \{\gamma b_\kappa : \gamma \in \mathbb{R}\}\Big) \subset \mathbb{R}^{|\tilde{x}^{(2)}|} \times \mathbb{R}^{|\mathcal{K}|}.$$

Thus, at most one coordinate of $\tilde{\gamma}_i$ is nonzero. This restricted support is compatible with the standard BLP share integrals and therefore defines an admissible random-coefficients distribution for the demand system.

We consider the model with primitives $(\beta, \alpha, \phi, \tilde{F}_{\beta^{(2)},\gamma})$, where β and α are the mean coefficients on $x^{(1)}_{jt}$ and p_{jt}, $\phi = (\phi_1, \ldots, \phi_{|\mathcal{K}|})'$ denotes the distribution of information types, and $\tilde{F}_{\beta^{(2)},\gamma}$ is the joint distribution of the random coefficients $(\beta^{(2)}_i, \gamma_i)$. We show that these primitives are identified from market-level data on $(s_t, x^{(1)}_t, \tilde{x}^{(2)}_t, \tilde{g}^e_t, p_t, z_t)$, where

$z_t = \{z_{jt}\}_{j \in \mathcal{J}_t}$ denotes instruments excluded from the unobserved demand shocks.

The argument proceeds in three steps. In Step 1, we establish identification of the demand system $\sigma(\cdot)$ and the mean utilities δ using the results of Berry and Haile (2014). In Step 2, given the identified demand system, we apply Wang (2023) to identify the induced random-coefficients distribution $F_{\beta^{(2)},\tilde{\gamma}}$. In Step 3, exploiting the support restriction $\tilde{\gamma}_i = \gamma_i b_{\kappa(i)}$, we recover the structural primitives $(\beta, \alpha, \phi, \tilde{F}_{\beta^{(2)},\gamma})$ from the identified objects.

Step 1 (Identification of the demand system and mean utilities).

Berry and Haile (2014) study identification in a general random utility model with market-level data. Under Proposition 1, our specification (10) is a special case of their framework. Their results establish identification of (i) the mean utilities δ_{jt} (and hence the demand shocks ξ_{jt}) under invertibility and instrument exogeneity, and (ii) the demand system $\sigma(\cdot)$ under a completeness condition. However, they do not identify the distribution of random coefficients, which we address in Step 2 following Wang (2023).

We follow Berry and Haile (2014) and consider market-level data where a market t is characterized by $(\mathcal{J}_t, \chi_t)$, with $\mathcal{J}_t$ the set of products and $\chi_t = (x_t^{(1)}, x_t^{(2)}, p_t, \xi_t)$ collecting observed characteristics, prices, and unobserved demand shocks. Conditional on χ_t, the utilities $\{u_{ijt}^e\}_{j \in \mathcal{J}_t}$ are i.i.d. across consumers i with joint distribution $F_{u^e}(\cdot|\chi_t)$. Market shares are given by

$$s_{jt} = \sigma_j(\chi_t) = \Pr\left(\arg\max_{k \in \mathcal{J}_t} u_{ikt}^e = j \mid \chi_t \right), \qquad j = 0, \ldots, J,$$

and we denote the demand system by $\sigma(\chi_t) = (\sigma_1(\chi_t), \ldots, \sigma_J(\chi_t))'$.

Let the mean utility index be

$$\delta_{jt} = x_{jt}^{(1)\prime}\beta + \xi_{jt}, \qquad \delta_t = (\delta_{1t}, \ldots, \delta_{Jt})'.$$

The outside option utility is normalized to zero and we impose $\mathbb{E}[\xi_{jt}] = 0$ for $j = 1, \ldots, J$.

We adapt the main assumptions in Berry and Haile (2014) as follows.

Assumption BH1 (index restriction).

$$F_{u^e}(\cdot|\chi_t) = F_{u^e}(\cdot|\delta_t, x_t^{(2)}, p_t).$$

Assumption BH2 (connected substitutes). Goods $(0, 1, \ldots, J)$ are connected substitutes in $-p_t$ and in δ_t.

Assumption BH3 (IV exogeneity).

$$\mathbb{E}[\xi_{jt} \mid z_t, x_t^{(1)}, x_t^{(2)}] = 0 \quad \text{for } j = 1, \ldots, J.$$

Assumption BH4 (completeness). For any measurable function $h(s_t, p_t; x_t^{(2)}, F_{\beta^{(2)},\tilde{\gamma}})$ with finite expectation,

$$\mathbb{E}[h(s_t, p_t; x_t^{(2)}, F_{\beta^{(2)},\tilde{\gamma}}) \mid z_t, x_t^{(1)}, x_t^{(2)}] = 0$$

implies $h(s_t, p_t; x_t^{(2)}, F_{\beta^{(2)},\tilde{\gamma}}) = 0$ almost surely.

Assumptions BH1–BH2 ensure invertibility of the demand system. Assumption BH3 is the instrument exogeneity restriction. In our context it requires that the predicted attributes $\tilde{g}_t^e$ be conditionally mean independent of the demand shocks ξ_{jt}. Assumption BH4 is a completeness condition ensuring injectivity of the nonparametric IV mapping.

Proposition 3. *Under Proposition 1 and Assumptions BH1–BH4, the demand system $\sigma(\cdot)$ and the mean utilities δ_t in model* (10) *are identified on the support $\mathcal{X}$.*

Proof. Under Proposition 1, model (10) is a random utility model satisfying the framework of Berry and Haile (2014). The result therefore follows directly from their Theorem 1. ■

Step 2 (Identification of the random-coefficients distribution). Given the identified demand system $\sigma(\cdot)$, we apply the results of Wang (2023) to identify the distribution of random coefficients $F_{\beta^{(2)},\tilde{\gamma}}$.

We impose the following assumptions adapted from Wang (2023).

Assumption W1 (logit kernel). Taste shocks $(\epsilon_{i0t}, \epsilon_{i1t}, \ldots, \epsilon_{iJt})$ are i.i.d. Type-I extreme value with scale normalized to 1.

Assumption W2 (exogenous taste heterogeneity). The distribution of random coefficients $(\beta_i^{(2)}, \gamma_i)$ is independent of $(x_{jt}^{(1)}, x_{jt}^{(2)}, p_{jt}, \xi_{jt})_{j \in \mathcal{J}_t \setminus \{0\}}$ and $(\epsilon_{ijt})_{j \in \mathcal{J}_t}$.

Assumption W3 (rank condition).[18] There exists markets t such that the matrix $x_t^{(2)\prime}$ has full column rank.

In our setting the random-coefficient regressors are

$$x_{jt}^{(2)} = (\tilde{x}_{jt}^{(2)}, \tilde{g}_{jt}^e).$$

Thus Assumption W3 requires that the columns of $\tilde{g}_t^e$ are not perfectly collinear across infor-

[18] It is incorporated into Assumption 7 in the main text.

mation types. This condition is analogous to Assumption 5 in the benchmark identification analysis.

Proposition 4. *Under Proposition 3 and Assumptions W1–W3, the random-coefficients distribution* $F_{\beta^{(2)},\tilde{\gamma}}$ *in model* (10) *is identified.*

Proof. By Proposition 3, the demand system $\sigma_j(\delta, p; x_t^{(2)}, F_{\beta^{(2)},\tilde{\gamma}})$ is identified on the support $\mathcal{X}$ as a function of δ on an open set $\mathcal{D} \subset \mathbb{R}^J$. This verifies Condition 1 of Wang (2023). Assumption W3 corresponds to their rank condition (Condition 2). The result therefore follows from Theorem 1 of Wang (2023). ■

Step 3 (Identification of the structural primitives).

We now recover the economically meaningful primitives $(\beta, \alpha, \phi, \tilde{F}_{\beta^{(2)},\gamma})$ from the identified objects $(\delta, F_{\beta^{(2)},\tilde{\gamma}})$.

Identification of (β, α)**.** From Step 1 the mean utilities satisfy

$$\delta_{jt} = x_{jt}^{(1)\prime}\beta - \alpha p_{jt} + \xi_{jt}.$$

Under Assumption BH3 we have the moment conditions

$$\mathbb{E}[\xi_{jt}(\beta, \alpha) \mid z_t, x_t^{(1)}, x_t^{(2)}] = 0.$$

Let Z_t denote functions of $(z_t, x_t^{(1)})$. These imply unconditional moments

$$\mathbb{E}[\xi_t(\beta, \alpha) \otimes Z_t] = 0.$$

We impose the following rank condition.

Assumption 9. *For some market* t*, the Jacobian*

$$C_t(\beta_0, \alpha_0) = \nabla_{\beta,\alpha}\, \mathbb{E}[\xi_t(\beta, \alpha) \otimes Z_t]\big|_{(\beta,\alpha)=(\beta_0,\alpha_0)}$$

has full column rank.

Assumption 9 ensures that the moment conditions uniquely determine (β_0, α_0) locally, establishing identification of the mean utility parameters.

Identification of $(\phi, \tilde{F}_{\beta^{(2)},\gamma})$**.** Step 2 identifies the joint distribution $F_{\beta^{(2)},\tilde{\gamma}}$ where

$$\tilde{\gamma}_i = \gamma_i b_{\kappa(i)}.$$

Hence $\tilde{\gamma}_i$ is supported on the union of the coordinate axes, and at most one coordinate of $\tilde{\gamma}_i$ is nonzero.

For each type κ, define the off-origin axis event

$$E_\kappa = \{\tilde{\gamma} \in \mathbb{R}^{|\mathcal{K}|} : \tilde{\gamma}_\kappa \neq 0,\ \tilde{\gamma}_{\kappa'} = 0 \text{ for all } \kappa' \neq \kappa\}.$$

Under the maintained independence Assumption 7 $(\beta_i^{(2)}, \gamma_i) \perp \kappa(i)$ with $\Pr(\kappa(i) = \kappa) = \phi_\kappa$, we have

$$\Pr(\tilde{\gamma}_i \in E_\kappa) = \phi_\kappa \Pr(\gamma_i \neq 0).$$

Therefore, provided $\Pr(\gamma_i \neq 0) > 0$,

$$\phi_\kappa = \frac{\Pr(\tilde{\gamma}_i \in E_\kappa)}{\sum_{\ell=1}^{|\mathcal{K}|} \Pr(\tilde{\gamma}_i \in E_\ell)}.$$

Thus the information-type distribution ϕ is identified from the relative off-origin masses on the coordinate axes.

Fix any κ with $\phi_\kappa > 0$ and note that on E_κ the active coordinate satisfies $\tilde{\gamma}_{i,\kappa} = \gamma_i$. Hence, for any Borel set $\mathcal{B} \subseteq \mathbb{R}^{|x^{(2)}|} \times (\mathbb{R} \setminus \{0\})$,

$$\Pr\left((\beta_i^{(2)}, \gamma_i) \in \mathcal{B}\right) = \frac{\Pr\big((\beta_i^{(2)}, \tilde{\gamma}_{i,\kappa}) \in \mathcal{B},\ \tilde{\gamma}_i \in E_\kappa\big)}{\phi_\kappa}.$$

The off-origin-γ_i part of $(\beta_i^{(2)}, \gamma_i) \mid \kappa$ is therefore identified from the distribution of the nonzero coordinate along any axis. The mass of γ_i at zero is recovered from

$$\Pr(\gamma_i = 0) = \Pr(\tilde{\gamma}_i = 0),$$

since $\tilde{\gamma}_i = \gamma_i b_{\kappa(i)}$. Thus, the conditional distribution $\tilde{F}_{(\beta^{(2)},\gamma)|\kappa}$ is identified. Finally, under Assumption 7, the unconditional distribution $\tilde{F}_{\beta^{(2)},\gamma}$ is identified.

We summarize step 3 in the following proposition:

Proposition 5. *Under the maintained assumptions, the primitives* $(\beta, \alpha, \phi, \tilde{F}_{\beta^{(2)},\gamma})$ *are identified from* $(\delta, F_{\beta^{(2)},\tilde{\gamma}})$.

Combining Steps 1–3 yields the following result.

Theorem 4. *Under Propositions 1, 3, 4, and 5, the structural primitives* $(\beta, \alpha, \phi, \tilde{F}_{\beta^{(2)},\gamma})$ *in model* (10) *are identified from market-level data on* $(s_t, x_t^{(1)}, \tilde{x}_t^{(2)}, \tilde{g}_t^e, p_t, z_t)$.

A.4 Proof of Theorem 3

This appendix proves Theorem 3. The result is algebraic: it follows from the finite-mixture structure of market shares and does not require the selection/inference assumptions used for implementation in Section 5.

Setup. Fix a market t with outside option 0 and inside options $j \in \{1, \dots, J\}$. Let $\mathcal{K}_t$ denote a finite set of information types. Type $\kappa \in \mathcal{K}_t$ has logit choice probabilities

$$s_{\kappa jt}(\theta) = \frac{\exp\left(\delta_{jt}(\beta) + \gamma g^e_{\kappa jt}\right)}{1 + \sum_{j'=1}^{J} \exp\left(\delta_{j't}(\beta) + \gamma g^e_{\kappa j't}\right)}, \qquad s_{\kappa 0t}(\theta) = \frac{1}{1 + \sum_{j'=1}^{J} \exp\left(\delta_{j't}(\beta) + \gamma g^e_{\kappa j't}\right)}, \tag{23}$$

where $\theta = (\beta', \gamma)'$ and $\delta_{jt}(\beta) \equiv X'_{jt}\beta + \xi_{jt}$.

Let $\phi_\kappa \geq 0$ denote the population share of type κ, with $\sum_{\kappa \in \mathcal{K}_t} \phi_\kappa = 1$. Observed market shares satisfy the finite-mixture restriction

$$s_{jt} = \sum_{\kappa \in \mathcal{K}_t} \phi_\kappa s_{\kappa jt}(\theta), \qquad s_{0t} = \sum_{\kappa \in \mathcal{K}_t} \phi_\kappa s_{\kappa 0t}(\theta). \tag{24}$$

Assume $s_{0t} > 0$.

For each type, the share ratio simplifies to

$$\frac{s_{\kappa jt}(\theta)}{s_{\kappa 0t}(\theta)} = \exp\left(\delta_{jt}(\beta) + \gamma g^e_{\kappa jt}\right). \tag{25}$$

Proof of Theorem 3. Fix (j, t). Using (24), write the observed share ratio as

$$\begin{aligned} \frac{s_{jt}}{s_{0t}} &= \frac{\sum_{\kappa \in \mathcal{K}_t} \phi_\kappa s_{\kappa jt}(\theta)}{\sum_{\kappa \in \mathcal{K}_t} \phi_\kappa s_{\kappa 0t}(\theta)} \\ &= \sum_{\kappa \in \mathcal{K}_t} \underbrace{\frac{\phi_\kappa s_{\kappa 0t}(\theta)}{\sum_{\kappa' \in \mathcal{K}_t} \phi_{\kappa'} s_{\kappa' 0t}(\theta)}}_{=:\, \omega_{\kappa t}(\theta)} \cdot \frac{s_{\kappa jt}(\theta)}{s_{\kappa 0t}(\theta)}. \end{aligned}$$

Because $s_{0t} > 0$, the weights $\omega_{\kappa t}(\theta)$ are well-defined, satisfy $\omega_{\kappa t}(\theta) \geq 0$ for all κ, and sum to one. Hence $\frac{s_{jt}}{s_{0t}}$ is a convex combination of the type-specific ratios $\left\{\frac{s_{\kappa jt}(\theta)}{s_{\kappa 0t}(\theta)}\right\}_{\kappa \in \mathcal{K}_t}$, which implies

$$\min_{\kappa \in \mathcal{K}_t} \left\{\frac{s_{\kappa jt}(\theta)}{s_{\kappa 0t}(\theta)}\right\} \leq \frac{s_{jt}}{s_{0t}} \leq \max_{\kappa \in \mathcal{K}_t} \left\{\frac{s_{\kappa jt}(\theta)}{s_{\kappa 0t}(\theta)}\right\}. \tag{26}$$

Substituting the logit share-ratio identity (25) into (26) and applying $\log(\cdot)$ (monotone) yields the population bounds

$$\delta_{jt}(\beta) + \min_{\kappa \in \mathcal{K}_t}\{\gamma g^e_{\kappa jt}\} \leq \log \frac{s_{jt}}{s_{0t}} \leq \delta_{jt}(\beta) + \max_{\kappa \in \mathcal{K}_t}\{\gamma g^e_{\kappa jt}\}. \tag{27}$$

Rearranging gives exactly the population inequalities stated in the Theorem (equivalently, the conditional moments in (12)–(13) evaluated at the true θ_0 are nonnegative). In particular, letting $\theta_0 = (\beta_0', \gamma_0)'$ denote the true preference parameters generating the mixture shares, the inequalities hold at θ_0, which implies $\theta_0 \in \Theta_0$ by definition of the identified set. ■

Appendix B Revisiting DM18

This appendix summarizes the model in DM18 and documents our replication and extensions. We first restate the export-decision model and the information variables considered in DM18. We then describe how we (i) implement the finite-mixture specification over information types and estimate it by maximum likelihood, and (ii) implement the extremal moment-inequality approach.

B.1 Model

Firm i decides whether to export to destination j in year t:

$$d_{ijt} \equiv \mathbb{1}\{\mathbb{E}\left[\pi_{ijt} \mid \mathcal{I}_{ijt}\right] \geq 0\},$$

where $\mathcal{I}_{ijt}$ denotes the information set used by firm i in year t to form predictions about the potential export revenue r_{ijt}. Profits take the form

$$\pi_{ijt} = \eta^{-1} r_{ijt} - f_{ijt},$$

where η is the demand elasticity (calibrated to $\eta = 5$ in DM18) and f_{ijt} is a fixed exporting cost, specified as:

$$f_{ijt} = \beta_0 + \beta_1 \, dist_j + \nu_{ijt},$$

where $dist_j$ denotes distance to destination j and ν_{ijt} is a demand shock. The export decision can be written as

$$d_{ijt} \equiv \mathbb{1}\Big\{\eta^{-1}\mathbb{E}[r_{ijt} \mid \mathcal{I}_{ijt}] - \beta_0 - \beta_1 \, dist_j - \nu_{ijt} \geq 0\Big\}.$$

The demand shock follows a normal distribution $v_{ijt}|(\mathcal{I}_{ijt}, dist_j) \overset{\text{iid}}{\sim} N(0, \sigma^2)$. The probability that firm i exports to a destination j in year t writes as:

$$\Pr(d_{ijt} = 1|\mathcal{I}_{ijt}, dist_j; \theta) = \Phi\left[\sigma^{-1}\left(\eta^{-1}\mathbb{E}\left[r_{ijt}|\mathcal{I}_{ijt}\right] - \beta_0 - \beta_1 dist_j\right)\right], \tag{28}$$

where $\Phi(\cdot)$ denotes the CDF of the standard normal distribution and the parameter of interest is $\theta = (\beta_0, \beta_1, \sigma)'$.

The export revenue r_{ijt} takes the role of the uncertain attribute g_{ijt} in our model. However, we do not observe the revenue r_{ijt} of a firm i if she does not export to the destination j in year t. To deal with the issue, DM18 further specifies the potential revenue as a function of the observed domestic revenue r_{iht} of firm i in year t:

$$r_{ijt} = \alpha_{jt} r_{iht} + e_{ijt},$$

where the revenue shifter α_{jt} is a sufficient statistic of how destination-specific supply and demand factors rescale the value of the domestic revenue to the (counterfactual) potential export revenue r_{ijt}, and e_{ijt} is the error term. We follow their approach and estimate the revenue shifter using data on firms that have exported to the destination j in year t. Under the assumption that $\mathbb{E}_{jt}\left[e_{ijt}|\mathcal{I}_{ijt}, r_{iht}, f_{ijt}\right] = 0$. $\alpha_{jt} r_{iht}$ equals the uncertain attribute g_{ijt} in our model.

DM18 discuss the role of five observed information candidates: distance $dist_j$, last year's domestic revenue r_{iht-1}, last year's aggregate revenue R_{jt-1} from all firms that have exported to the destination j, last year's revenue shifter α_{jt-1}, and the number of firms N_{jt-1} that have exported to the destination j in year $t-1$. They assume all exporters know the first three information variables and construct the minimal information set $\mathcal{I}^m_{ijt} = \{dist_j, r_{iht-1}, R_{jt-1}\}$. We retain their minimal information set and use α_{jt-1} and N_{jt-1} to construct additional information types. We construct $K_3 = 4$ information types, indexed by $\kappa = A, B, C, D$, where the information set of each type is $\mathcal{W}_A = \{dist_j, r_{iht-1}, R_{jt-1}\}$, $\mathcal{W}_B = \{dist_j, r_{iht-1}, R_{jt-1}, \alpha_{jt-1}\}$, $\mathcal{W}_C = \{dist_j, r_{iht-1}, R_{jt-1}, N_{jt-1}\}$, and $\mathcal{W}_D = \{dist_j, r_{iht-1}, R_{jt-1}, \alpha_{jt-1}, N_{jt-1}\}$, respectively.

B.2 Estimation with Finite Mixture Approach

In each sector and year t, we observe the export decisions of all firms d_{ijt}, their domestic revenue r_{iht}, the export revenue r_{ijt} of firms who have exported to the destination j, and a list of information candidates $\mathcal{K}_{ijt} = (dist_j, r_{iht-1}, R_{jt-1}, \alpha_{jt-1}, N_{jt-1})$.

We separately estimate the export decision in each sector with the maximum likelihood. In this context, the likelihood function (11) becomes

$$\mathcal{L}(\Theta|d,\mathcal{K},dist)=\prod_{it}\prod_{j}\left[\Pr(d_{ijt}=1|\mathcal{K}_{ijt},dist_j;\Theta)\right]^{d_{ijt}}\left[1-\Pr(d_{ijt}=1|\mathcal{K}_{ijt},dist_j;\Theta)\right]^{1-d_{ijt}},$$

and the choice probability $\Pr(d_{ijt}=1|\mathcal{K}_{ijt},dist_j;\Theta)$ is specified as

$$\Pr(d_{ijt}=1|\mathcal{K}_{ijt},dist_j;\Theta)=\sum_{\kappa}^{K_3}\phi_\kappa\Phi\left[\sigma^{-1}\left(\eta^{-1}\mathbb{E}\left[\alpha_{jt}r_{iht}|\mathcal{W}_\kappa\right]-\beta_0-\beta_1 dist_j\right)\right], \tag{29}$$

where the fraction parameters satisfy the constraints $\sum_{\kappa}^{K_3}\phi_\kappa=1;\phi_\kappa\in[0,1]$ and the parameter vector is $\Theta=(\beta_0,\beta_1,\sigma,\phi_A,\phi_B,\phi_C,\phi_D)'$.

When bringing the likelihood function (29) to the data, we need to calculate the type-specific expectations of the potential export revenue $\mathbb{E}\left[\alpha_{jt}r_{iht}|\mathcal{W}_\kappa\right]$. We calculate the conditional expectation with a non-linear exponential specification following DM18. For instance, the expectation of the potential export revenue in information type A, i.e., the minimal information type, is

$$\mathbb{E}\left[\alpha_{jt}r_{iht}|\mathcal{W}_A\right]=\exp\left(\gamma_{A1}\log(dist_j)+\gamma_{A2}\log(r_{iht-1})+\gamma_{A3}\log(R_{jt-1})\right), \tag{30}$$

where we first run a non-linear regression of the ex-post export revenue $\alpha_{jt}r_{iht}$ on information variables $\log(dist_j),\log(r_{iht-1}),\log(R_{jt-1})$ and obtain the estimated coefficients $\hat{\gamma}_{A1},\hat{\gamma}_{A2},\hat{\gamma}_{A3}$. Then we predict the unobserved expectation as

$$\hat{\mathbb{E}}\left[\alpha_{jt}r_{iht}|\mathcal{W}_A\right]=\exp(\hat{\gamma}_{A1}\log(dist_j)+\hat{\gamma}_{A2}\log(r_{iht-1})+\hat{\gamma}_{A3}\log(R_{jt-1})).$$

We repeat this calculation for each information type κ. In implementation, we omit an intercept term from this auxiliary prediction regression. When an intercept is included, the fitted values of $\widehat{\mathbb{E}}[\alpha_{jt}r_{iht}\mid\mathcal{W}_\kappa]$ are nearly degenerate in our sample (close to zero across types), which makes the resulting type-specific predictions empirically indistinguishable and prevents the mixture model from separating information types. Dropping the intercept restores meaningful variation in predicted expectations across information types, which is the variation exploited in the finite-mixture estimation.

B.3 Estimation with Moment Inequalities

Following DM18, we estimate the standard deviation $\sigma > 0$ of the demand shock ν_{ijt}, and we abstract from any unobserved product attribute ξ_{jt}. We assume that ν_{ijt} follows an EVT1 distribution instead of a normal distribution. This allows us to directly adapt our inequality specification in Equation (15) as follows:

$$\mathbf{m}^{\tilde{\kappa}}_{\max}(Z_{ijt};\theta) \equiv \mathbb{E}\left[\left(-\log\frac{s_{ijt}}{1-s_{ijt}} + \sigma^{-1}\left(\eta^{-1}\mathbb{E}[r_{ijt} \mid \mathcal{I}^{\tilde{\kappa}}_{ijt}] - \beta_0 - \beta_1 dist_j\right)\right) \cdot Z^S_{\max\, ijt}\middle| Z_{ijt}\right],$$

$$\mathbf{m}^{\tilde{\kappa}}_{\min}(Z_{ijt};\theta) \equiv \mathbb{E}\left[\left(\log\frac{s_{ijt}}{1-s_{ijt}} - \sigma^{-1}\left(\eta^{-1}\mathbb{E}[r_{ijt} \mid \mathcal{I}^{\tilde{\kappa}}_{ijt}] - \beta_0 - \beta_1 dist_j\right)\right) \cdot Z^S_{\min\, ijt}\middle| Z_{ijt}\right],$$

with $Z_{ijt} = (\text{constant}, dist_j, \mathcal{I}_{\tilde{\kappa},ijt})$ and $\mathbb{E}\left[r_{ijt} \mid \mathcal{I}_{\tilde{\kappa},ijt}\right] = \alpha_{jt} r_{iht} - e_{\tilde{\kappa}jt}$ denoting the predicted export revenue of an observed information type $\tilde{\kappa}$.

Since our approach relies on observing choice probabilities (market shares), we group firms with similar revenues $\alpha_{jt} r_{iht}$ into 20 quantile groups for each destination j in year t. The group-specific export probabilities are $s_{jgt} = \frac{\sum_{i \in g} d_{ijt}}{N_g}$, where N_g is the number of firms in revenue group g. We lose information by aggregating s_{ijt} to s_{gjt}. Consequently, some observations lead to moment inequalities that are too weak to reject any candidate parameter values. We exclude these data points from the estimation sample. Alternatively, we could specify an alternative parameter grid, but we prefer to evaluate the inequalities on the same grid as in DM18.

We define the minimal information type as the observed type $\tilde{\kappa}$. Because only about 11% of firms in the sample (chemical sector) are exporters, the estimated group-specific export probabilities s_{gjt} are typically very small and estimated from relatively sparse cells. This leads to limited variation in the lower tail of the estimated probabilities and, after the log-odds transformation, provides little identifying variation for the min-side inequalities. As a result, in practice the min-side inequalities are much less informative than the max-side inequalities.

To address this issue, we use asymmetric selection thresholds in Assumption 8. Specifically, after estimating the predicted export revenues $\hat{\mathbb{E}}\left[r_{ijt} \mid \mathcal{I}^m_{ijt}\right]$ for the minimal information type $\mathcal{I}^m_{ijt}$, we select extreme predictions below the 5-th percentile, i.e., $Z^S_{\min\, jt} = \mathbb{1}\left\{\hat{\mathbb{E}}\left[r_{ijt} \mid \mathcal{I}^m_{ijt}\right] \leq q_{0.05}\right\}$, to construct the min-side moments. We use predictions above the 93-th percentile, i.e., $Z^S_{\max\, jt} = \mathbb{1}\left\{\hat{\mathbb{E}}\left[r_{ijt} \mid \mathcal{I}^m_{ijt}\right] \geq q_{0.93}\right\}$, to construct the max-side moments.

Appendix C Additional Tables

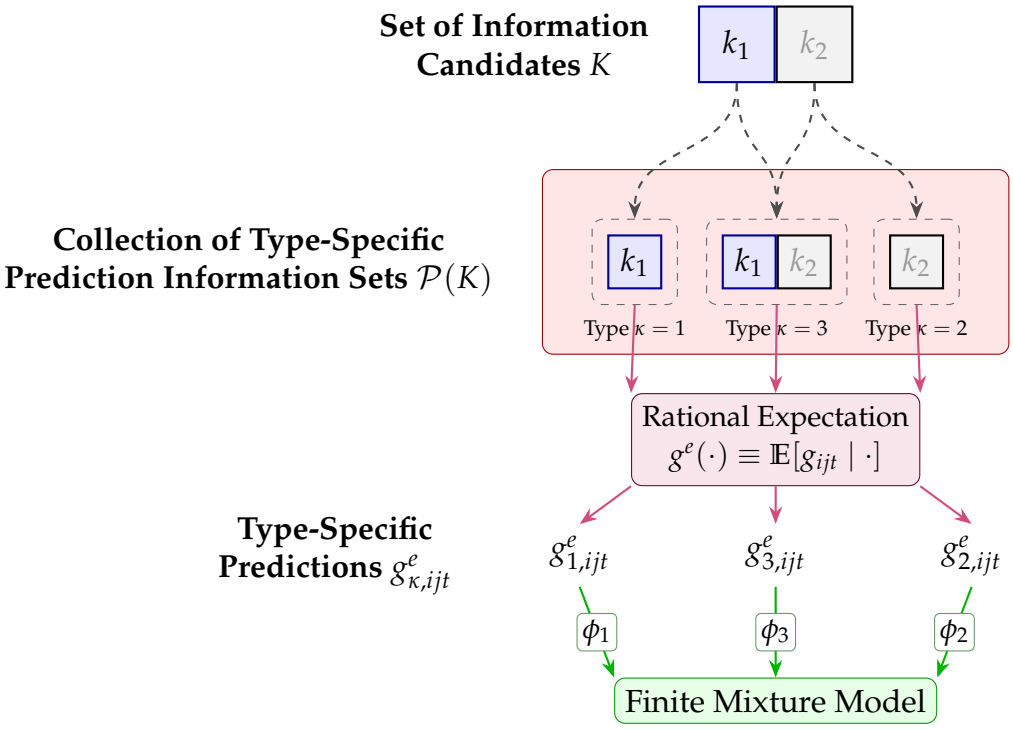
Table 5: Demand Estimation with Unobserved Information Factors in the Two-type (B, C included) Simulation

		Regular IV			Chamberlain's IV		
param.	true	est.	st. err.	bias	est.	st. err.	bias
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
β_0	1	2.58	(4.160)	1.58	1.00	(0.084)	0.00
β_x	1	1.08	(1.111)	0.08	1.00	(0.053)	0.00
α	1.5	1.62	(1.121)	-0.12	1.50	(0.045)	-0.00
ϕ_A	0	0.09	(0.095)	0.09	2.31E-04	(0.001)	0.00
ϕ_B	0.4	0.63	(0.189)	0.23	0.40	(0.018)	0.00
ϕ_C	0.6	0.29	(0.157)	-0.31	0.60	(0.019)	-0.00
γ	-1.5	-2.93	(0.867)	-1.43	-1.51	(0.027)	-0.01

Notes. All estimates are from 1,000 Monte Carlo simulations and the DGP includes two information types $\kappa = B, C$. Regular instruments are the intercept, product characteristics x_{jt}, prices p_{jt}, cost shifter z_{jt}, and two information variables k_{1jt}, k_{2jt} with k_{1jt} observed and k_{2jt} estimated by a linear factor model. Chamberlain's IV replaces the latter two information variables in the instrument set.

Appendix D Additional Figures

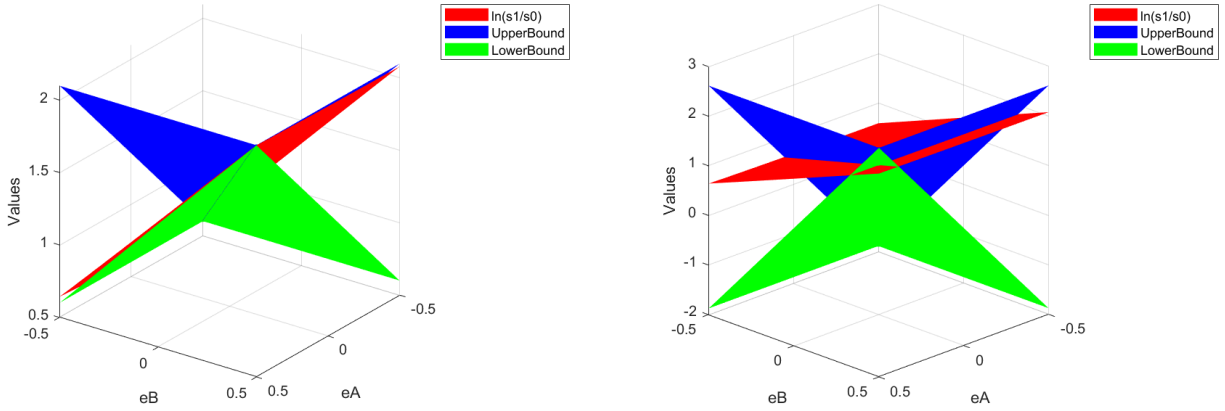
Figure A1: Visual representation of the information structure construction.



Notes. This figure illustrates the process of constructing type-specific prediction information sets from the set of information candidates consisting of one observed information candidate k_1 and a latent factor k_2. The power set $\mathcal{P}(K)$ generates distinct information types, which are then mapped through $g^e(\cdot)$ to form rational expectations.

Figure A2: Graphical Illustration of Inequalities

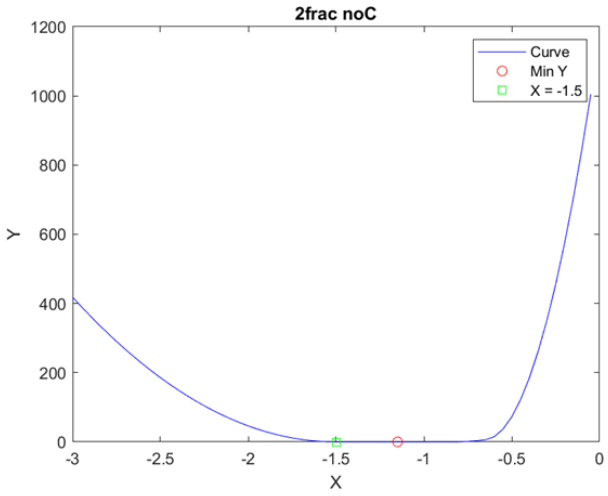
(a) Under True Value of $\gamma = -1.5$ (b) Under False Value of $\gamma = -4.5$



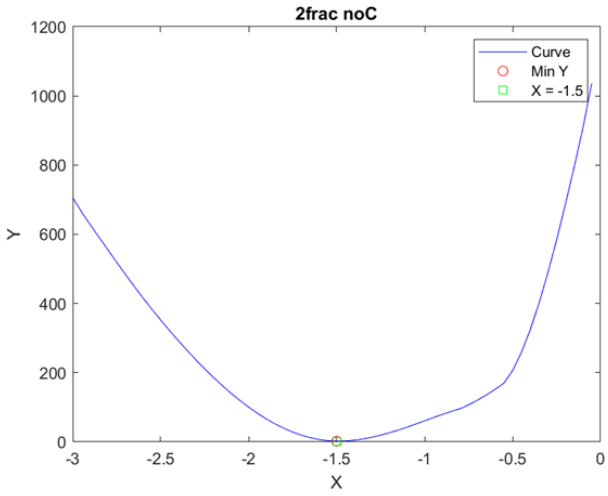
Notes. This figure illustrates the power of our inequalities (14) in identifying the correct values of the parameter γ. The right-hand side of (14) is plotted in blue, while the left-hand side is in green. The observed log market share ratio is plotted in red. Given the true parameter value $\gamma = -1.5$, plot A2a shows that the max side in blue always bounds the market share ratio from above while the min side in green bounds the data from below. However, given a wrong parameter value $\gamma = -4.5$, our inequalities are violated, which will then rule out such wrong parameter value from the identified set, as depicted in plot A2b.

Figure A3: Comparison of GMM Objective Function with vs. without Chamberlain IVs

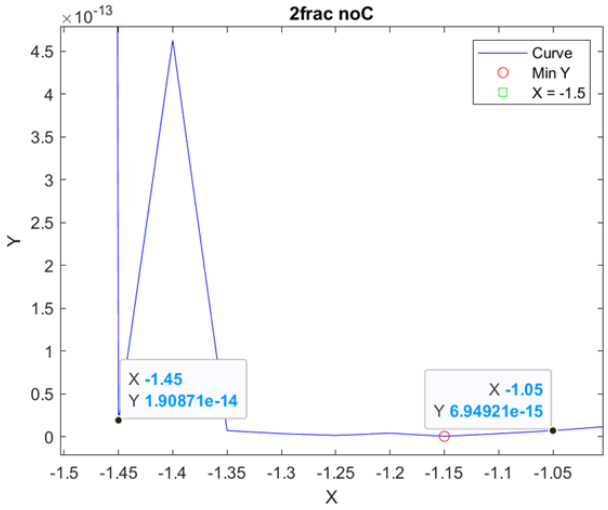
(a) Regular IVs: Gobal View



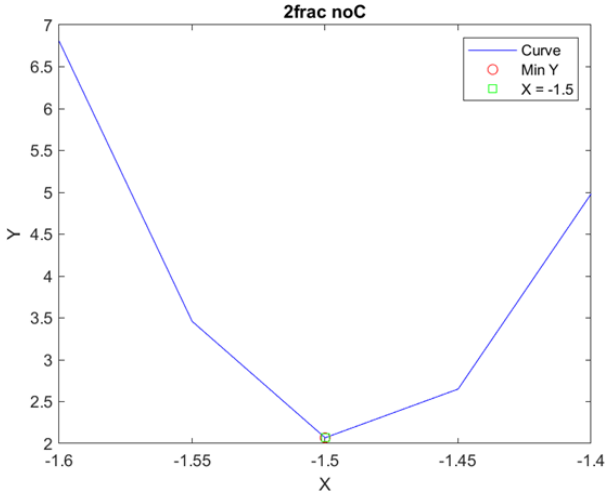
(c) Chamberlain IVs: Gobal View



(b) Regular IVs: Local View

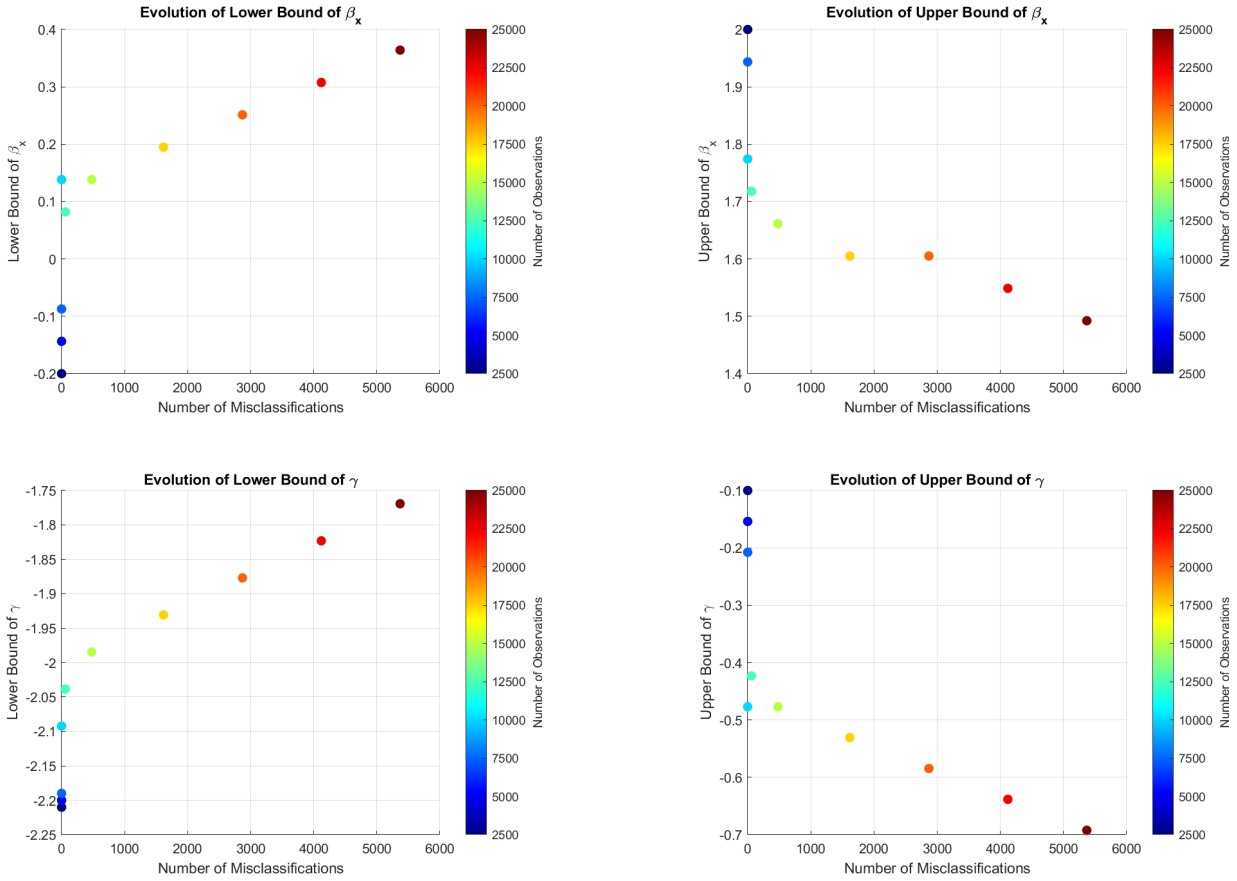


(d) Chamberlain IVs: Local View



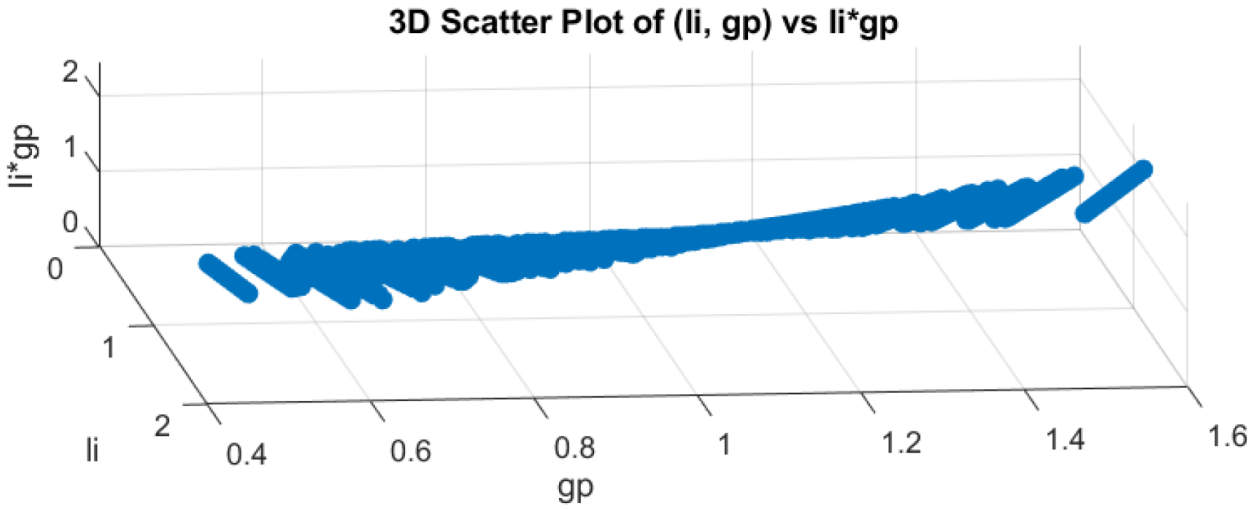
Notes. This figure displays the GMM objective function using two types of instruments: standard IVs in panels (a) and (b) on the left, and Chamberlain IVs in panels (c) and (d) on the right. The top panels (a, c) illustrate the global shape of the objective function, while the bottom panels (b, d) focus on the region where the function attains its minimum. The horizontal axis shows the estimated parameter γ; the vertical axis shows the value of the objective function. The true parameter value, $\gamma = -1.5$, is marked by the green box, and the red circle indicates the objective function's minimum. We observe that the regular instruments lead to a problematic objective function with multiple local minima. However, the GMM objective function becomes well-behaved when using Chamberlain's instruments, further supports the effectiveness of Chamberlain's instruments.

Figure A4: Estimated bounds for β_x and γ across different selection thresholds ($\tilde{\kappa} = A$)



Notes. This figure plots the estimated bounds for each parameter β_x and γ across different thresholds. The y-axis corresponds to the value of the estimated bound. The x-axis corresponds to the number of misclassification in the estimation sample. The color indicates the sample size. We observed that, as the sample size increases, the confidence interval becomes more precise (the lower bound increases while the upper bound decreases).

Figure A5: Non-linear Relationship between Information Variables and Fuel Costs



Notes. This figure plots the distribution of the fuel costs (g_{jkt}, i.e., "li*gp" on the z-axis) as a function of two information variables–fuel prices (f_{kt}, i.e., "gp" on the x-axis) and fuel efficiency (e_{jkt}, i.e., "li" on the y-axis). The fuel costs are defined as $g_{jkt} := e_{jkt} \cdot f_{kt}$.